
Energy and chemical change

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You have probably seen a fire burning or burnt fuel for warmth or cooking or light. A fire burning is one of the most noticeable examples of a chemical reaction that produces a lot of energy.

All chemical reactions involve energy changes. In some reactions, we are able to observe these energy changes as either an increase or a decrease in the overall energy of the system. In some reactions we see this as a change in the temperature. In other reactions we can observe this change when a reaction starts to give off light or when a reaction will only work after light is shone on it.



The study of energy changes (particularly heat) in chemical reactions is known as chemical thermodynamics. This is also sometimes called thermochemistry.

▶ See video: 2447 at www.everythingscience.co.za

Key Mathematics Concepts

- Graphs — Mathematics, Grade 10, Functions and graphs
- Equations — Mathematics, Grade 10, Equations and inequalities
- Units and unit conversions — Physical Sciences, Grade 10, Science skills

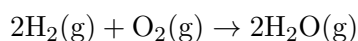
12.1 Energy changes in chemical reactions

ESBQJ

What causes the energy changes in chemical reactions?

ESBQK

When a chemical reaction occurs, bonds in the reactants *break*, while new bonds *form* in the product. The following example explains this. Hydrogen reacts with oxygen to form water, according to the following equation:



In this reaction, the bond between the two hydrogen atoms in the H_2 molecule will *break*, as will the bond between the oxygen atoms in the O_2 molecule. New bonds will *form* between the two hydrogen atoms and the single oxygen atom in the water molecule that is formed as the product.

For bonds to *break*, energy must be *absorbed*. When new bonds *form*, energy is *released*. The energy that is needed to break a bond is called the **bond energy** or **bond dissociation energy**. Bond energies are measured in units of $\text{kJ}\cdot\text{mol}^{-1}$.

DEFINITION: Bond energy

Bond energy is a measure of bond strength in a chemical bond. It is the amount of energy (in $\text{kJ}\cdot\text{mol}^{-1}$) that is needed to break the chemical bond between two atoms.

Remember when we discussed bonding (chapter 3) we used the following energy diagram:

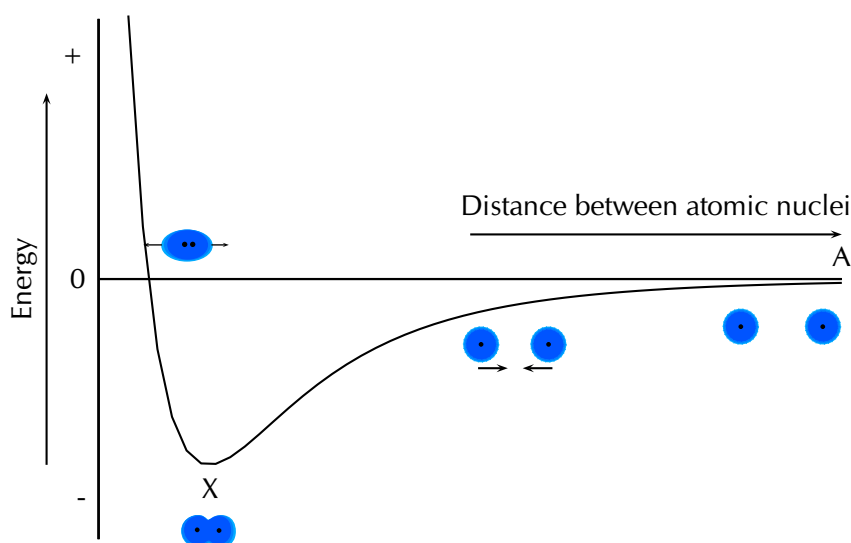
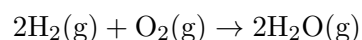


Figure 12.1: Graph showing the changes in energy that take place as the distance between two atoms changes.

We can use this diagram to understand why bond breaking requires energy and bond making releases energy. Point X on the diagram is at the lowest energy. When a bond breaks, the atoms move apart and the distance between them increases (i.e. the atom moves to the right on the x -axis or from point X to point A). Looking at the diagram we see that when this happens, the energy increases (i.e. the energy at point A is greater than the energy at point X). So when a bond breaks energy is needed.

When a bond forms the atoms move closer together and the distance between them decreases (i.e. the atom moves to the left on the x -axis or from point A to point X). Looking at the diagram we see that when this happens, the energy decreases (i.e. the energy at point X is less than the energy at point A). So when a bond forms energy is released.

Looking at the example of hydrogen reacting with oxygen to form water:



We see that energy is needed to break the bonds in the hydrogen molecule and to break the bonds in the oxygen molecule. And we also see that energy is released when hydrogen and oxygen bond to form water. When we look at the entire reaction and consider both bond breaking and bond forming we need to look at the **enthalpy** of the system.

DEFINITION: *Enthalpy*

Enthalpy is a measure of the total energy of a chemical system for a given pressure, and is given the symbol H .

As we learn about exothermic and endothermic reactions we will see more on the concept of enthalpy.

TIP

A chemical system is a closed system that contains only the reactants and products involved in the reaction.

TIP

Δ is read as delta and means a change in. You may recall this symbol from physics.

Exothermic and endothermic reactions

ESBQM

In some reactions, the energy that must be *absorbed* to break the bonds in the reactants, is less than the energy that is *released* when the new bonds of the products are formed. This means that in the overall reaction, energy is *released* as either heat or light. This type of reaction is called an **exothermic** reaction.

DEFINITION: *Exothermic reaction*

An exothermic reaction is one that releases energy in the form of heat or light.

Another way of describing an exothermic reaction is that it is one in which the energy of the products is less than the energy of the reactants, because energy has been released during the reaction. We can represent this using the following general formula:



In other reactions, the energy that must be *absorbed* to break the bonds in the reactants, is more than the energy that is *released* when the new bonds in the products are formed. This means that in the overall reaction, energy must be *absorbed* from the surroundings. This type of reaction is known as an **endothermic** reaction.

DEFINITION: *Endothermic reaction*

An endothermic reaction is one that absorbs energy in the form of heat or light.

Another way of describing an endothermic reaction is that it is one in which the energy of the products is greater than the energy of the reactants, because energy has been absorbed during the reaction. This can be represented by the following general formula:



The difference in energy (E) between the reactants and the products is known as the **heat of the reaction**. It is also sometimes referred to as the **enthalpy change** of the system. This is represented using ΔH

Formal experiment: Endothermic and exothermic reactions - part 1

Apparatus and materials:

You will need:

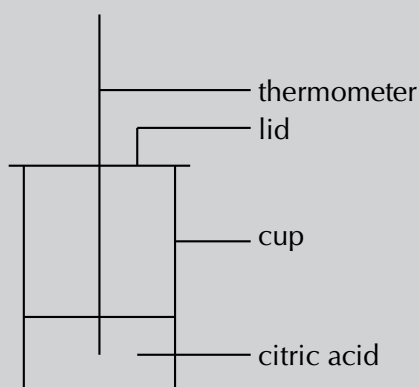
- citric acid
- sodium bicarbonate
- a polystyrene cup
- a lid for the cup

- thermometer
- glass stirring rod
- scissors

Note that citric acid is found in citrus fruits such as lemons. Sodium bicarbonate is actually bicarbonate of soda (baking soda), the baking ingredient that helps cakes to rise.

Method:

1. If your lid does not have a hole for a straw, then cut a small hole into the lid.
2. Pour some citric acid ($\text{C}_6\text{H}_8\text{O}_7$) into the polystyrene cup, cover the cup with its lid and record the temperature of the solution.
3. Stir in the sodium bicarbonate (NaHCO_3), then cover the cup again.
4. Immediately record the temperature, and then take a temperature reading every two minutes after that. Record your results.



The equation for the reaction that takes place is:



Results:

Time (mins)	0	2	4	6
Temperature ($^{\circ}\text{C}$)				

Plot your temperature results on a graph of time (x -axis) against temperature (y -axis).

Discussion and conclusion:

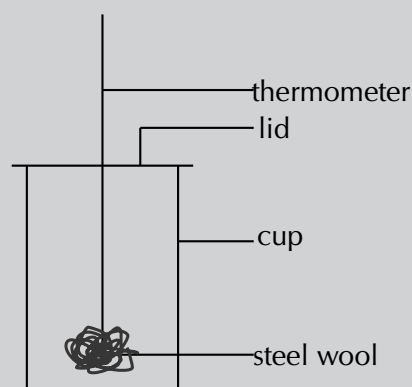
- What happens to the temperature during this reaction?
- Is this an exothermic or an endothermic reaction? (Was energy taken in or given out? Did the temperature increase or decrease?)
- Why was it important to keep the cup covered with a lid?

Apparatus and materials:

- Vinegar
- steel wool
- thermometer
- polystyrene cup and plastic lid (from previous experiment)

Method:

1. Put the thermometer through the plastic lid, cover the cup and record the temperature in the empty cup. You will need to leave the thermometer in the cup for about 5 minutes in order to get an accurate reading.
2. Soak a piece of steel wool in vinegar for about a minute. The vinegar removes the protective coating from the steel wool so that the metal is exposed to oxygen.
3. Take the thermometer out of the cup. Keep the thermometer through the hole of the lid.
4. After the steel wool has been in the vinegar, remove it and squeeze out any vinegar that is still on the wool. Wrap the steel wool around the thermometer and place it (still wrapped round the thermometer) back into the cup. The cup is automatically sealed when you do this because the thermometer is through the top of the lid.
5. Leave the steel wool in the cup for about 5 minutes and then record the temperature. Record your observations.



Results:

You should notice that the temperature *increases* when the steel wool is wrapped around the thermometer.

Conclusion:

The reaction between oxygen and the exposed metal in the steel wool is **exothermic**, which means that energy is released and the temperature increases.

TIP

Note that we are only discussing chemical changes (recall from grade 10 about physical and chemical changes). Physical changes can also be classified as exothermic or endothermic. When we are referring to physical change then we talk about exothermic or endothermic processes. Evaporation is an endothermic process while condensation is an exothermic process.

There are many examples of endothermic and exothermic reactions that occur around us all the time. The following are just a few examples.

1. Endothermic reactions**• Photosynthesis**

Photosynthesis is the chemical reaction that takes place in green plants, which uses energy from the sun to change carbon dioxide and water into food that the plant needs to survive, and which other organisms (such as humans and other animals) can eat so that they too can survive. The equation for this reaction is:



Photosynthesis is an endothermic reaction. Energy in the form of sunlight is absorbed during the reaction.

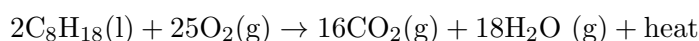
• The thermal decomposition of limestone

In industry, the breakdown of limestone into quicklime and carbon dioxide is very important. Quicklime can be used to make steel from iron and also to neutralise soils that are too acid. However, the limestone must be heated in a kiln (oven) at a temperature of over 900 °C before the decomposition reaction will take place. The equation for the reaction is shown below:

**2. Exothermic reactions****• Combustion reactions**

The burning of fuel is an example of a combustion reaction, and we as humans rely heavily on this process for our energy requirements. The following equations describe the combustion of a hydrocarbon such as *petrol* (C_8H_{18}):

fuel + oxygen → heat + water + carbon dioxide



This is why we burn fuels (such as paraffin, coal, propane and butane) for energy, because the chemical changes that take place during the reaction release huge amounts of energy, which we then use for things like power and electricity. You should also note that *carbon dioxide* is produced during this reaction. The chemical reaction that takes place when fuels burn has both positive and negative consequences. Although we benefit from heat, power and electricity the carbon dioxide that is produced has a negative impact on the environment.

• Respiration

Respiration is the chemical reaction that happens in our bodies to produce energy for our cells. The equation below describes what happens during this reaction:



In the reaction above, glucose (a type of carbohydrate in the food we eat) reacts with oxygen from the air that we breathe in, to form carbon dioxide

FACT

Lightsticks or glowsticks are used by divers, campers, and for decoration and fun. A lightstick is a plastic tube with a glass vial inside it. To activate a lightstick, you bend the plastic stick, which breaks the glass vial. This allows the chemicals that are inside the glass to mix with the chemicals in the plastic tube. These two chemicals react and release energy. Another part of a lightstick is a fluorescent dye which changes this energy into light, causing the lightstick to glow! This is known as phosphorescence or chemiluminescence.



(which we breathe out), water and energy. The energy that is produced allows the cell to carry out its functions efficiently. Can you see now why you must eat food to get energy? It is not the food itself that provides you with energy, but the exothermic reaction that takes place when compounds within the food react with the oxygen you have breathed in!

▶ See video: 2448 at www.everythingscience.co.za

▶ See video: 2449 at www.everythingscience.co.za

Exercise 12 – 1: Exothermic and endothermic reactions 1

1. State whether energy is taken in or released in each of the following situations:
 - a) The bond between hydrogen and chlorine in a molecule of hydrogen chloride breaks.
 - b) A bond is formed between hydrogen and fluorine to form a molecule of hydrogen fluoride.
 - c) A molecule of nitrogen (N_2) is formed.
 - d) A molecule of carbon monoxide breaks apart.
2. State whether the following descriptions are used to describe an endothermic or an exothermic reaction:
 - a) Reactants react to give products and energy.
 - b) The energy that must be absorbed to break the bonds in the reactants is greater than the energy that is released when the products form.
 - c) The energy of the products is found to be greater than the energy of the reactants for this type of reaction.
 - d) Heat or light must be absorbed from the surroundings before this type of reaction takes place.

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2c. 244J 2d. 244K



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12.2 Exothermic and endothermic reactions ESBQP

The heat of reaction

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TIP

ΔH has been calculated for many different reactions and so instead of saying that ΔH is positive or negative, we can look up the value of ΔH for the reaction and use that value instead.

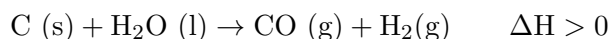
The **heat of the reaction** is represented by the symbol ΔH , where:

$$\Delta H = E_{\text{prod}} - E_{\text{react}}$$

- In an *exothermic* reaction, ΔH is less than zero because the energy of the reactants is greater than the energy of the products. Energy is released in the reaction. For example:



- In an *endothermic* reaction, ΔH is greater than zero because the energy of the reactants is less than the energy of the products. Energy is absorbed in the reaction. For example:



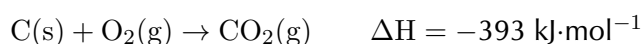
Some of the information relating to exothermic and endothermic reactions is summarised in Table 12.1.

Type of reaction	Exothermic	Endothermic
Energy absorbed or released	Released	Absorbed
Relative energy of reactants and products	Energy of reactants greater than energy of product	Energy of reactants less than energy of product
Sign of ΔH	Negative (i.e. < 0)	Positive (i.e. > 0)

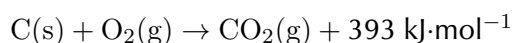
Table 12.1: A comparison of exothermic and endothermic reactions.

Writing equations using ΔH

There are two ways to write the heat of the reaction in an equation. For the exothermic reaction $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$, we can write:



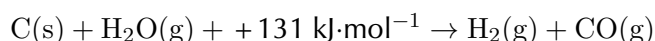
or



For the endothermic reaction, $\text{C}(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_2(\text{g}) + \text{CO}(\text{g})$, we can write:



or



The **units** for ΔH are $\text{kJ}\cdot\text{mol}^{-1}$. In other words, the ΔH value gives the amount of energy that is absorbed or released per mole of product that is formed. Units can also be written as kJ, which then gives the total amount of energy that is released or absorbed when the product forms.

The energy changes during exothermic and endothermic reactions can be plotted on a graph:

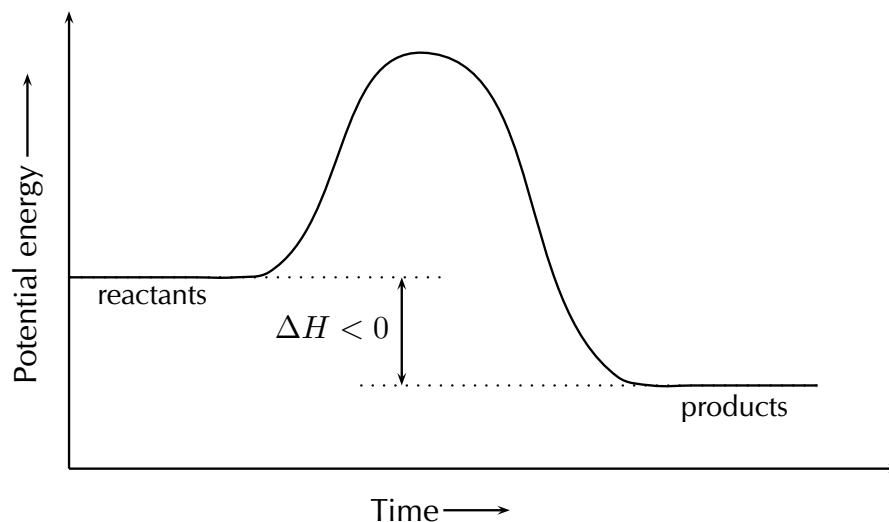


Figure 12.2: The energy changes that take place during an exothermic reaction.

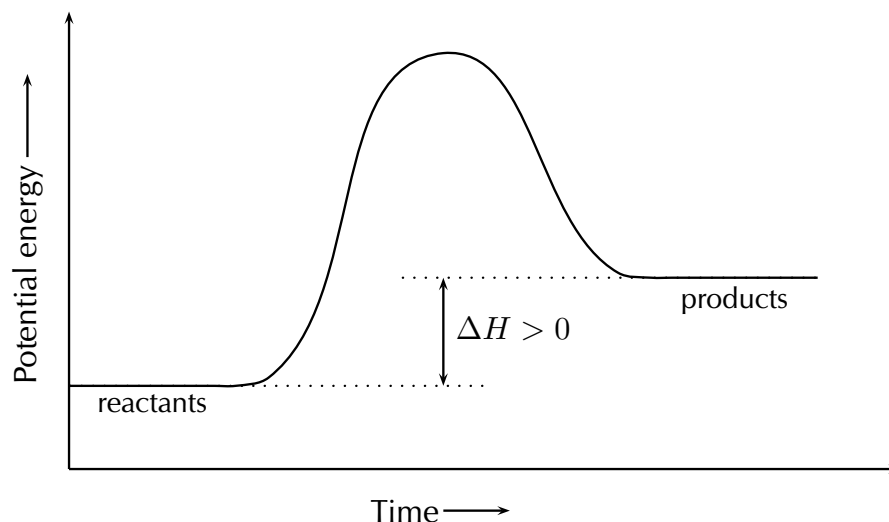


Figure 12.3: The energy changes that take place during an endothermic reaction.

We will explain shortly why we draw these graphs with a curve rather than simply drawing a straight line from the reactants energy to the products energy.

Investigation: Endothermic and exothermic reactions

Aim:

To investigate exothermic and endothermic reactions.

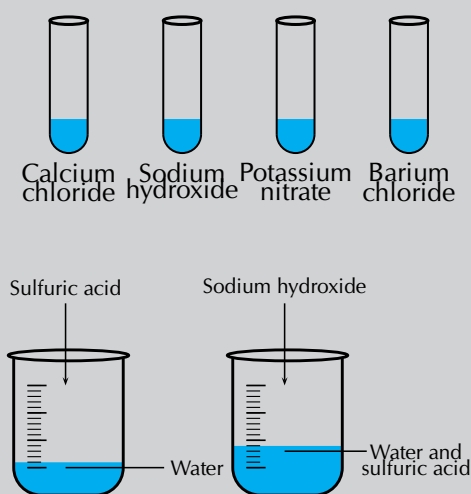
Apparatus and materials:

- Approximately 2 g of calcium chloride (CaCl_2)
- Approximately 2 g of sodium hydroxide (NaOH)
- Approximately 2 g of potassium nitrate (KNO_3)
- Approximately 2 g of barium chloride (BaCl_2)
- concentrated sulphuric acid (H_2SO_4) (Be careful, this can cause serious burns)
- 5 test tubes
- thermometer

WARNING!

When working with concentrated sulfuric acid always wear gloves and safety glasses. Always work in a well ventilated room or in a fume cupboard.

Method:



1. Dissolve about 1 g of each of the following substances in 5-10 cm^3 of water in a test tube: CaCl_2 , NaOH , KNO_3 and BaCl_2 .
2. Observe whether the reaction is endothermic or exothermic, either by feeling whether the side of the test tube gets hot or cold, or using a thermometer.
3. Dilute 3 cm^3 of concentrated H_2SO_4 in 10 cm^3 of water in the fifth test tube and observe whether the temperature changes.

WARNING!

Remember to always add the acid to the water.

4. Wait a few minutes and then carefully add NaOH to the diluted H_2SO_4 . Observe any temperature (energy) changes.

Results:

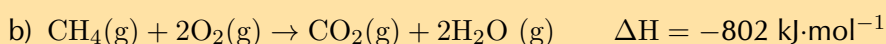
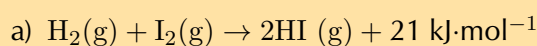
Record which of the above reactions are endothermic and which are exothermic.

Exothermic reactions	Endothermic reactions

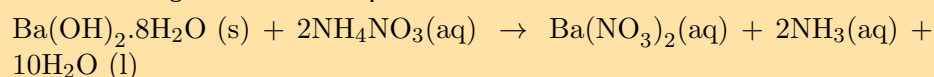
- When BaCl_2 and KNO_3 dissolve in water, they take in heat from the surroundings. The dissolution of these salts is **endothermic**.
- When CaCl_2 and NaOH dissolve in water, heat is released. The process is **exothermic**.
- The reaction of H_2SO_4 and NaOH is also **exothermic**.

Exercise 12 – 2: Endothermic and exothermic reactions

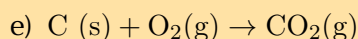
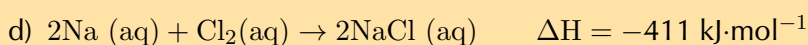
1. In each of the following reactions, say whether the reaction is endothermic or exothermic, and give a reason for your answer. Draw the resulting energy graph for each reaction.



- c) The following reaction takes place in a flask:



Within a few minutes, the temperature of the flask drops by approximately 20°C .



2. For each of the following descriptions, say whether the process is endothermic or exothermic and give a reason for your answer.

- a) evaporation
- b) the combustion reaction in a car engine
- c) bomb explosions
- d) melting ice
- e) digestion of food
- f) condensation

3. When you add water to acid the resulting solution splashes up. The beaker also gets very hot. Explain why.

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2b. 244T 2c. 244V 2d. 244W 2e. 244X 2f. 244Y 3. 244Z



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TIP

It is important to realise that even though exothermic reactions release energy they still need a small amount of energy to start the reaction.

12.3 Activation energy and the activated complex ESBQR

If you take a match and just hold it or wave it around in the air, the match will not light. You have to strike the match against the side of the box. All chemical reactions need something that makes them start going.

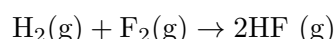
Chemical reactions will not take place until the system has some minimum amount of energy added to it. This energy is called the **activation energy**.

DEFINITION: Activation energy

Activation energy is the minimum amount of energy that is needed to start a chemical reaction.

Recall from earlier that we drew graphs for the energy changes in exothermic and endothermic reactions. We can now add some information to these graphs. This will also explain why we draw these graphs with a curve rather than using a straight line from the reactants energy to the products energy.

We will start by looking at exothermic reactions. We will use:



as an example of an exothermic reaction.

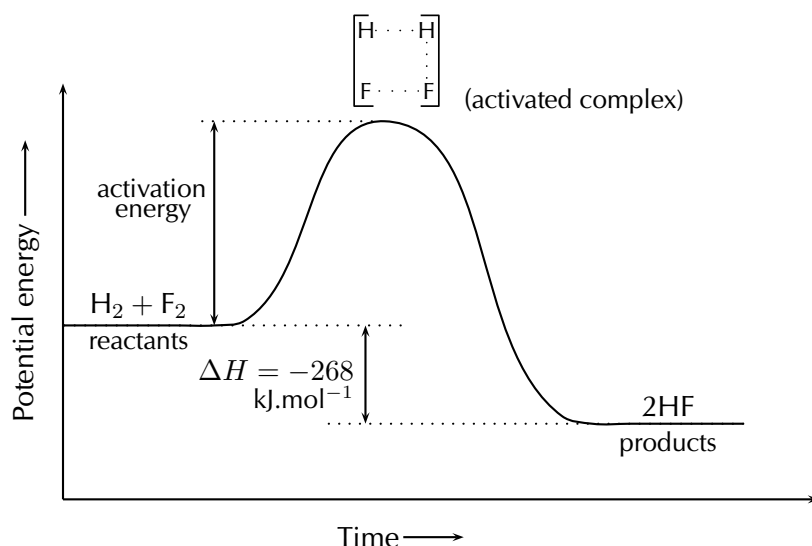


Figure 12.4: The energy changes that take place during an exothermic reaction.

TIP

The activation energy is the difference between the energy of the **reactants** and the maximum energy (i.e. the energy of the activated complex).

FACT

The reaction between H_2 and F_2 was considered by NASA (National Aeronautics and Space Administration) as a fuel system for rocket boosters because of the energy that is released during this exothermic reaction.

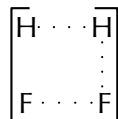
TIP

Enzymes and activation energy

An enzyme is a catalyst that helps to speed up the rate of a reaction by lowering the activation energy of a reaction. There are many enzymes in the human body, without which lots of important reactions would never take place. Cellular respiration is one example of a reaction that is catalysed by enzymes. You will learn more about catalysts in Grade 12.

The reaction between $\text{H}_2(\text{g})$ and $\text{F}_2(\text{g})$ (Figure 12.4) needs energy in order to proceed, and this is the activation energy. To form the product the bond between H and H in H_2 must break. The bond between F and F in F_2 must also break. A new bond between H and F must also form to make HF. The reactant bonds break at the same time that the product bonds form.

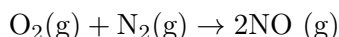
We can show this as:



This is called the **activated complex** or transition state. The activated complex lasts for only a *very short time*. After this short time one of two things will happen: the original bonds will reform, or the bonds are broken and a new product forms. In this example, the final product is HF and it has a lower energy than the reactants. The reaction is exothermic and ΔH is negative.

The activated complex is the complex that exists as the bonds in the products are forming and the bonds in the reactants are breaking. This complex exists for a very short period of time and is found when the energy of the system is at its maximum.

In endothermic reactions, the final products have a higher energy than the reactants. An energy diagram is shown below (Figure 12.5) for the endothermic reaction:



Notice that the activation energy for the endothermic reaction is much greater than for the exothermic reaction.

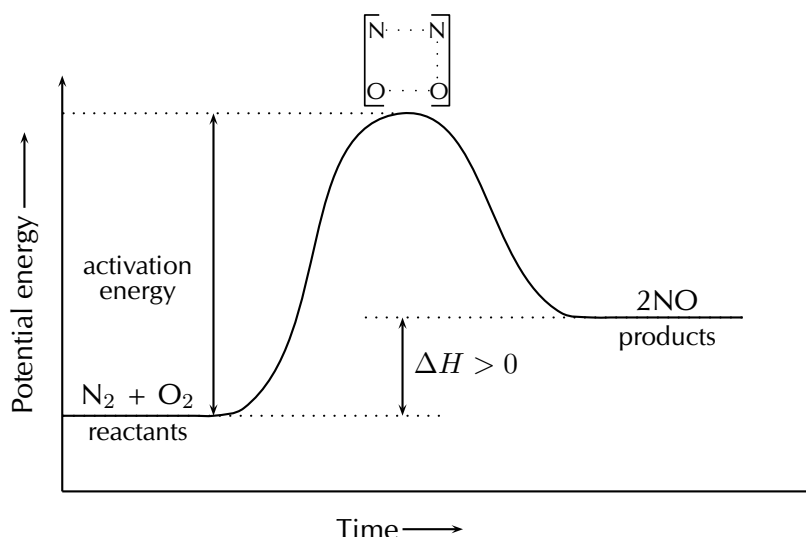


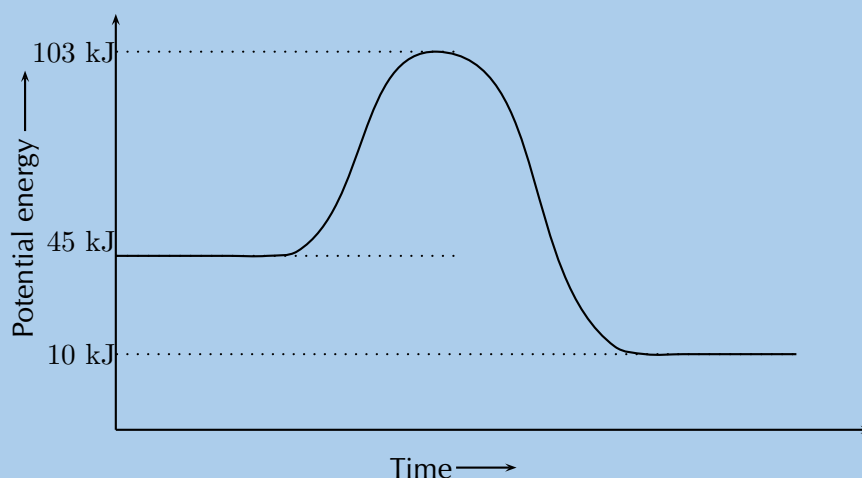
Figure 12.5: The energy changes that take place during an endothermic reaction.

It is because of this activation energy that we first need to show an increase in energy from the reactant to the activated complex and then a decrease in energy from the activated complex to the product. We show this on the energy graphs by drawing a curve from the energy of the reactants to the energy of the products.

Worked example 1: Activation energy

QUESTION

Refer to the graph below and then answer the questions that follow:



1. Calculate ΔH .
2. Is the reaction endothermic or exothermic and why?
3. Calculate the activation energy for this reaction.

SOLUTION

Step 1: Calculate ΔH

ΔH is found by subtracting the energy of the **reactants** from the energy of the **products**. We find the energy of the reactants and the products from the graph.

$$\begin{aligned}\Delta H &= \text{energy of products} - \text{energy of reactants} \\ &= 10 \text{ kJ} - 45 \text{ kJ} \\ &= -35 \text{ kJ}\end{aligned}$$

Step 2: Determine if this is exothermic or endothermic.

The reaction is exothermic since $\Delta H < 0$. We also note that the energy of the reactants is greater than the energy of the products.

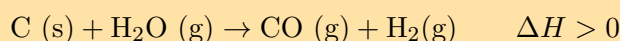
Step 3: Calculate the activation energy

The activation energy is found by subtracting the energy of the reactants from the energy of the activated complex. Again we can read the energy of the reactants and activated complex off the graph.

$$\begin{aligned}\text{activation energy} &= \text{energy of activated complex} - \text{energy of reactants} \\ &= 103 \text{ kJ} - 45 \text{ kJ} \\ &= 58 \text{ kJ}\end{aligned}$$

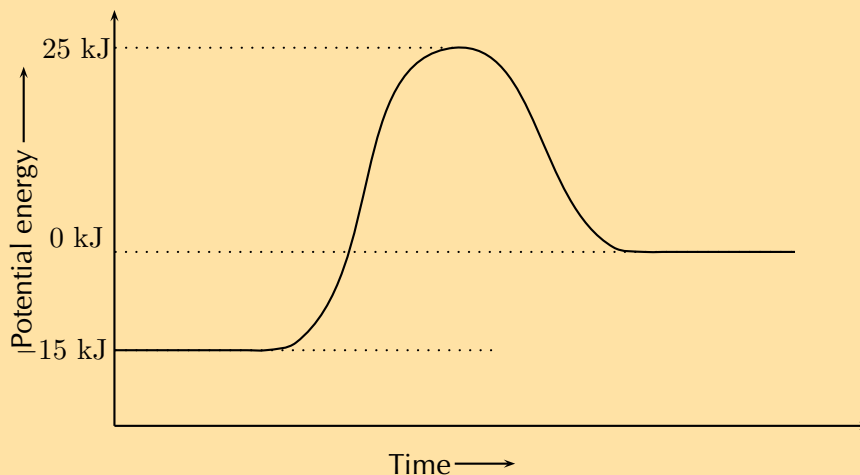
Exercise 12 – 3: Energy and reactions

1. Carbon reacts with water according to the following equation:



Is this reaction endothermic or exothermic? Give a reason for your answer.

2. Refer to the graph below and then answer the questions that follow:



- What is the energy of the reactants?
- What is the energy of the products?
- Calculate ΔH .
- What is the activation energy for this reaction?

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12.4 Chapter summary

ESBQS

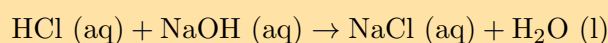
▶ See presentation: [2454](#) at www.everythingscience.co.za

- When a reaction occurs, bonds in the reactants break and new bonds in the products form. These changes involve **energy**.
- When bonds break, energy is **absorbed** and when new bonds form, energy is **released**.
- The **bond energy** is a measure of bond strength in a chemical bond. It is the amount of energy (in $\text{kJ}\cdot\text{mol}^{-1}$) that is needed to break the chemical bond between two atoms.
- Enthalpy** is a measure of the total energy of a chemical system for a given pressure and is given the symbol H .

- If the energy that is needed to *break* the bonds is less than the energy that is released when new bonds *form*, then the reaction is **exothermic**. The energy of the products is less than the energy of the reactants.
- An **exothermic** reaction is one that **releases energy** in the form of heat or light.
- If the energy that is needed to *break* the bonds is more than the energy that is released when new bonds *form*, then the reaction is **endothermic**. The energy of the products is greater than the energy of the reactants.
- An **endothermic** reaction is one that **absorbs energy** in the form of heat or light.
- Photosynthesis and the thermal decomposition of limestone are both examples of endothermic reactions.
- Combustion reactions and respiration are both examples of exothermic reactions.
- The difference in energy between the reactants and the product is called the **heat of reaction** and has the symbol ΔH .
- ΔH is calculated using: $\Delta H = E_{\text{prod}} - E_{\text{react}}$
- In an endothermic reaction, **ΔH is a positive number** (greater than 0). In an exothermic reaction, **ΔH will be negative** (less than 0).
- Chemical reactions will not take place until the system has some minimum amount of energy added to it.
- The **activation energy** is the minimum amount of energy that is needed to start a chemical reaction.
- The **activated complex** (or transition state) is the complex that exists as the bonds in the products are forming and the bonds in the reactants are breaking. This complex exists for a very short period of time and is found when the energy of the system is at its maximum.

Exercise 12 – 4:

- For each of the following, give one word or term for the description.
 - The minimum amount of energy that is needed for a reaction to proceed.
 - A measure of the bond strength in a chemical bond.
 - A type of reaction where ΔH is less than zero.
 - A type of reaction that requires heat or light to proceed.
- For the following reaction:



choose the correct statement from the list below.

- Energy is taken in when the new bonds in NaCl are formed.
- Energy is released when the bonds in HCl break.

- c) Energy is released when the bonds in H_2O form.
- d) Energy is released when the bonds in NaOH break.

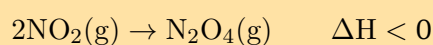
3. For the following reaction:



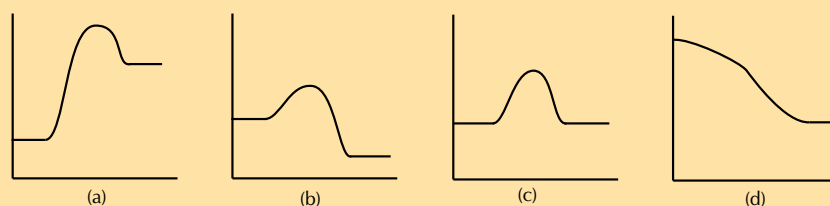
choose the correct statement from the list below.

- a) The energy of the reactants is less than the energy of the product.
- b) The energy of the product is less than the energy of the reactants.
- c) The reaction needs energy to occur.
- d) The overall energy of the system increases during the reaction.

4. Consider the following chemical reaction:



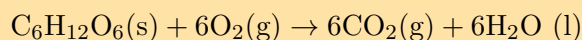
Which one of the following graphs best represents the changes in potential energy that take place during the production of N_2O_4 ?



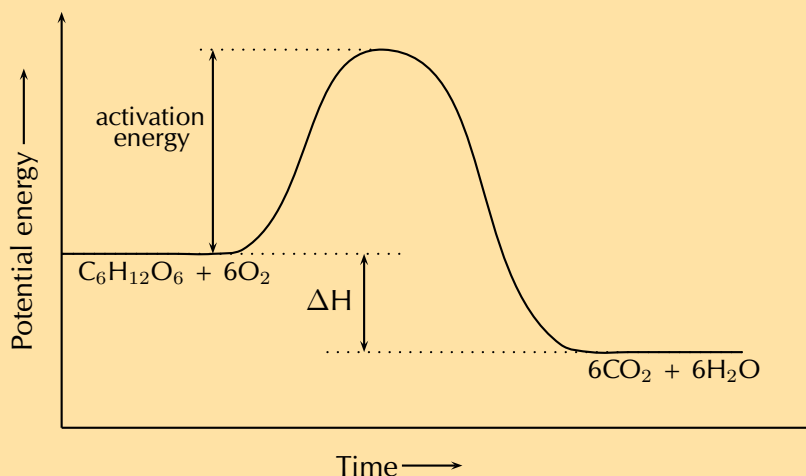
5. In each of the following reactions, say whether the reaction is endothermic or exothermic, and give a reason for your answer. Draw the resulting energy graph for each reaction.

- a) $\text{Fe}_2\text{O}_3(\text{s}) + 2\text{Al}(\text{s}) \rightarrow 2\text{Fe}(\text{s}) + \text{Al}_2\text{O}_3(\text{s}) + \text{heat}$
- b) $\text{NH}_4\text{Cl}(\text{s}) + \text{heat} \rightarrow \text{NH}_3(\text{g}) + \text{HCl}(\text{g})$

6. The cellular respiration reaction is catalysed by enzymes. The equation for the reaction is:



The change in potential energy during this reaction is shown below:



- a) Will the value of ΔH be positive or negative? Give a reason for your answer.
- b) Explain what is meant by activation energy.
- c) Glucose is one of the reactants in cellular respiration. What important chemical reaction produces glucose?
- d) Is the reaction in your answer above an endothermic or an exothermic one? Explain your answer.
- e) Draw the energy graph for the reaction that produces glucose.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1a. [2455](#) 1b. [2456](#) 1c. [2457](#) 1d. [2458](#) 2. [2459](#) 3. [245B](#)
4. [245C](#) 5a. [245D](#) 5b. [245F](#) 6. [245G](#)



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Types of reactions

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13.2	<i>Acid-base reactions</i>	443
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13.4	<i>Chapter summary</i>	465

All around you there are chemical reactions taking place. Green plants are photosynthesising, car engines are relying on the reaction between petrol and air and your body is performing many complex reactions. In this chapter we will look at two common types of reactions that can occur in the world around you and in the chemistry laboratory. These two types of reactions are acid-base reactions and redox reactions.

13.1 Acids and bases

ESBQT

What are acids and bases?

ESBQV

Activity: Household acids and bases

Look around your home and school and find examples of acids and bases. Remember that foods can also be acidic or basic.

Make a list of all the items you find. Why do you think they are acids or bases?

Some common acids and bases, and their chemical formulae, are shown in Table 13.1.

Acid	Formula	Base	Formula
Hydrochloric acid	HCl	Sodium hydroxide	NaOH
Sulfuric acid	H ₂ SO ₄	Potassium hydroxide	KOH
Sulfurous acid	H ₂ SO ₃	Sodium carbonate	Na ₂ CO ₃
Acetic (ethanoic) acid	CH ₃ COOH	Calcium hydroxide	Ca(OH) ₂
Carbonic acid	H ₂ CO ₃	Magnesium hydroxide	Mg(OH) ₂
Nitric acid	HNO ₃	Ammonia	NH ₃
Phosphoric acid	H ₃ PO ₄	Sodium bicarbonate	NaHCO ₃

Table 13.1: Some common acids and bases and their chemical formulae.

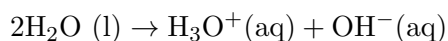
Most acids share certain characteristics, and most bases also share similar characteristics. It is important to be able to have a definition for acids and bases so that they can be correctly identified in reactions.

Defining acids and bases

ESBQW

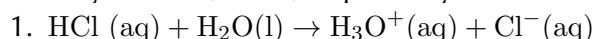
One of the first things that was noted about acids is that they have a sour taste. Bases were noted to have a soapy feel and a bitter taste. However you cannot go around tasting and feeling unknown substances since they may be harmful. Also when chemists started to write down chemical reactions more practical definitions were needed.

A number of definitions for acids and bases have developed over the years. One of the earliest was the **Arrhenius** definition. Arrhenius (1887) noticed that water dissociates (splits up) into hydronium (H_3O^+) and hydroxide (OH^-) ions according to the following equation:

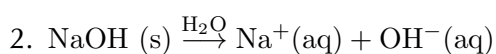


Arrhenius described an **acid** as a compound that increases the concentration of H_3O^+ ions in solution and a **base** as a compound that increases the concentration of OH^- ions in solution.

Look at the following examples showing the dissociation of hydrochloric acid and sodium hydroxide (a base) respectively:



Hydrochloric acid in water increases the concentration of H_3O^+ ions and is therefore an *acid*.



Sodium hydroxide in water increases the concentration of OH^- ions and is therefore a *base*.

Note that we write $\xrightarrow{\text{H}_2\text{O}}$ to indicate that water is needed for the dissociation.

However, this definition could only be used for acids and bases *in water*. Since there are many reactions which do not occur in water it was important to come up with a much broader definition for acids and bases.

In 1923, Lowry and Bronsted took the work of Arrhenius further to develop a broader definition for acids and bases. The **Bronsted-Lowry model** defines acids and bases in terms of their ability to donate or accept protons.

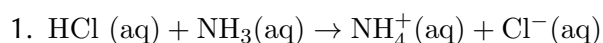
DEFINITION: *Acids*

A Bronsted-Lowry **acid** is a substance that gives away protons (hydrogen cations H^+), and is therefore called a **proton donor**.

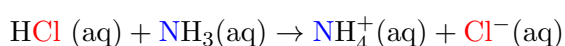
DEFINITION: *Bases*

A Bronsted-Lowry **base** is a substance that takes up protons (hydrogen cations H^+), and is therefore called a **proton acceptor**.

Below are some examples:



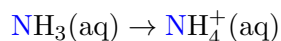
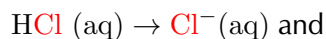
We highlight the chlorine and the nitrogen so that we can follow what happens to these two elements as they react. We do not highlight the hydrogen atoms as we are interested in how these change. This colour coding is simply to help you identify the parts of the reaction and does not represent any specific property of these elements.



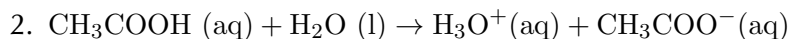
TIP

For more information on *dissociation*, refer to Grade 10 (chapter 18: reactions in aqueous solution).

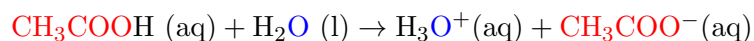
In order to decide which substance is a proton donor and which is a proton acceptor, we need to look at what happens to each reactant. The reaction can be broken down as follows:



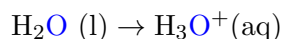
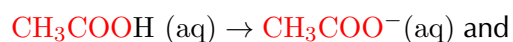
From these reactions, it is clear that HCl is a *proton donor* and is therefore an **acid**, and that NH₃ is a *proton acceptor* and is therefore a **base**.



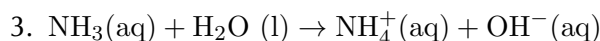
Again we highlight the parts of the reactants that we want to follow in this reaction:



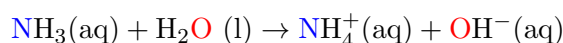
The reaction can be broken down as follows:



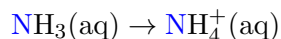
In this reaction, CH₃COOH (acetic acid or vinegar) is a proton donor and is therefore the **acid**. In this case, water acts as a **base** because it accepts a proton to form H₃O⁺.



Again we highlight the parts of the reactants that we want to follow in this reaction:



The reaction can be broken down as follows:



Water donates a proton and is therefore an **acid** in this reaction. Ammonia accepts the proton and is therefore the **base**.

Notice in these examples how we looked at the common elements to break the reaction into two parts. So in the first example we followed what happened to chlorine to see if it was part of the acid or the base. And we also followed nitrogen to see if it was part of the acid or the base. You should also notice how in the reaction for the acid there is one less hydrogen on the right hand side and in the reaction for the base there is an extra hydrogen on the right hand side.

Amphoteric substances

In examples 2 and 3 above we notice an interesting thing about water. In example 2 we find that water acts as a base (it accepts a proton). In example 3 however we see that water acts as an acid (it donates a proton)!

Depending on what water is reacting with it can either react as a base or as an acid. Water is said to be **amphoteric**. Water is not unique in this respect, several other substances are also amphoteric.

DEFINITION: Amphoteric

An amphoteric substance is one that can react as either an acid or base.

When we look just at Bronsted-Lowry acids and bases we can also talk about amphoteric substances which are a special type of amphoteric substances.

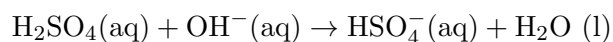
DEFINITION: *Amphiprotic*

An amphiprotic substance is one that can react as either a proton donor (Bronsted-Lowry acid) or as a proton acceptor (Bronsted-Lowry base). Examples of amphiprotic substances include water, hydrogen carbonate ion (HCO_3^-) and hydrogen sulfate ion (HSO_4^-).

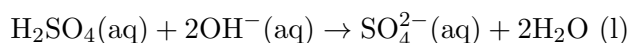
Note: You may also see the term **ampholyte** used to mean a substance that can act as both an acid and a base. This term is no longer in general use in chemistry.

Polyprotic acids [NOT IN CAPS]

A polyprotic (many protons) acid is an acid that has more than one proton that it can donate. For example sulfuric acid can donate one proton to form the hydrogen sulfate ion:



Or it can donate two protons to form the sulfate ion:



In this chapter we will mostly consider monoprotic acids (acids with only one proton to donate). If you do see a polyprotic acid in a reaction then write the resulting reaction equation with the acid donating all its protons.

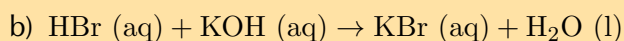
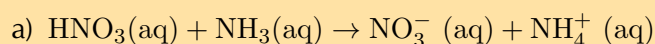
Some examples of polyprotic acids are:

H_2SO_4 , H_2SO_3 , H_2CO_3 and H_3PO_4 .

▶ See simulation: [245H](http://www.everythingscience.co.za/245H) at www.everythingscience.co.za

Exercise 13 – 1: Acids and bases

1. Identify the Bronsted-Lowry acid and the Bronsted-Lowry base in the following reactions:



2. a) Write a reaction equation to show HCO_3^- acting as an acid.

b) Write a reaction equation to show HCO_3^- acting as a base.

c) Compounds such as HCO_3^- are ...

TIP

Up to now you have looked at reactions as starting with the reactants and going to the products. For acids and bases we also need to consider what happens if we swop the reactants and the products around. This will help you understand conjugate acid-base pairs.

TIP

The word conjugate means coupled or connected.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1a. [245J](#) 1b. [245K](#) 2. [245M](#)



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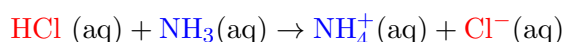


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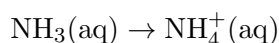
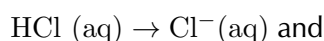
Conjugate acid-base pairs

ESBQX

Look at the reaction between hydrochloric acid and ammonia to form ammonium and chloride ions (again we have highlighted the different parts of the equation):

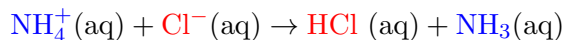


We look at what happens to each of the reactants in the reaction:



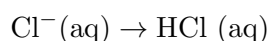
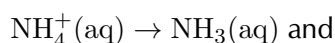
We see that HCl acts as the acid and NH₃ acts as the base.

But what if we actually had the following reaction:



This is the **same** reaction as the first one, but the products are now the reactants.

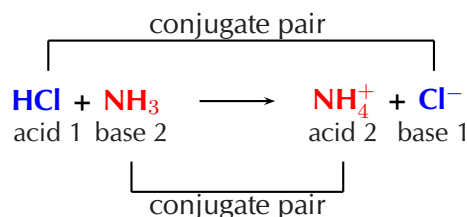
Now if we look at the what happens to each of the reactants we see the following:



We see that NH₄⁺ acts as the acid and Cl⁻ acts as the base.

When HCl (the acid) loses a proton it forms Cl⁻ (the base). And that when Cl⁻ (the base) gains a proton it forms HCl (the acid). We call these two species a **conjugate acid-base pair**. Similarly NH₃ and NH₄⁺ form a conjugate acid-base pair.

We can represent this as:



Activity: Conjugate acid-base pairs

Using the common acids and bases in Table 13.1, pick an acid and a base from the list. Write a chemical equation for the reaction of these two compounds.

Now identify the conjugate acid-base pairs in your chosen reaction. Compare your results to those of your classmates.

TIP

In chemistry the word salt does not mean the white substance that you sprinkle on your food (this white substance is a salt, but not the only salt). A salt (to chemists) is a product of an acid-base reaction and is made up of the cation from the base and the anion from the acid.

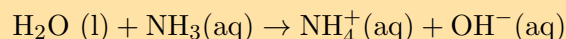
▶ See video: 245N at www.everythingscience.co.za

Exercise 13 – 2: Acids and bases

1. In each of the following reactions, label the conjugate acid-base pairs.

- a) $\text{H}_2\text{SO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{HSO}_4^-(\text{aq})$
- b) $\text{NH}_4^+(\text{aq}) + \text{F}^-(\text{aq}) \rightarrow \text{HF}(\text{aq}) + \text{NH}_3(\text{aq})$
- c) $\text{H}_2\text{O}(\text{l}) + \text{CH}_3\text{COO}^-(\text{aq}) \rightarrow \text{CH}_3\text{COOH}(\text{aq}) + \text{OH}^-(\text{aq})$
- d) $\text{H}_2\text{SO}_4(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{HCl}(\text{aq}) + \text{HSO}_4^-(\text{aq})$

2. Given the following reaction:



- a) Write down which reactant is the base and which is the acid.
- b) Label the conjugate acid-base pairs.
- c) In your own words explain what is meant by the term conjugate acid-base pair.

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1a. 245P 1b. 245Q 1c. 245R 1d. 245S 2. 245T



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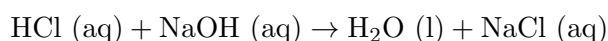
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13.2 Acid-base reactions

ESBQY

The reaction between an acid and a base is known as a **neutralisation** reaction. Often when an acid and base react a salt and water will be formed. We will look at a few examples of acid-base reactions.

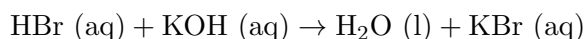
1. Hydrochloric acid reacts with sodium hydroxide to form sodium chloride (a salt) and water. Sodium chloride is made up of Na^+ cations from the base (NaOH) and Cl^- anions from the acid (HCl).



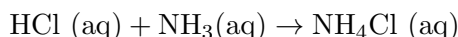
FACT

Bee stings are acidic and have a pH between 5 and 5.5. They can be soothed by using substances such as bicarbonate of soda and milk of magnesia. Both bases help to neutralise the acidic bee sting and relieve some of the itchiness!

2. Hydrogen bromide reacts with potassium hydroxide to form potassium bromide (a salt) and water. Potassium bromide is made up of K^+ cations from the base (KOH) and Br^- anions from the acid (HBr).



3. Hydrochloric acid reacts with ammonia to form ammonium chloride (a salt). Ammonium chloride is made up of NH_4^+ cations from the base (NH_3) and Cl^- anions from the acid (HCl).



You should notice that in the first two examples, the base contained OH^- ions, and therefore the products were a *salt* and *water*. NaCl (table salt) and KBr are both salts. In the third example, NH_3 also acts as a base, despite not having OH^- ions. A salt is still formed as the only product, but no water is produced.

It is important to realise how useful these neutralisation reactions are. Below are some examples:

- **Domestic uses**

Calcium oxide (CaO) is a base (all metal oxides are bases) that is put on soil that is too acidic. Powdered limestone (CaCO_3) can also be used but its action is much slower and less effective. These substances can also be used on a larger scale in farming and in rivers.

Limestone (white stone or calcium carbonate) is used in pit latrines (or long drops). The limestone is a base that helps to neutralise the acidic waste.

- **Biological uses**

Acids in the stomach (e.g. hydrochloric acid) play an important role in helping to digest food. However, when a person has a stomach ulcer, or when there is too much acid in the stomach, these acids can cause a lot of pain. **Antacids** are taken to neutralise the acids so that they don't burn as much. Antacids are bases which neutralise the acid. Examples of antacids are aluminium hydroxide, magnesium hydroxide ("milk of magnesia") and sodium bicarbonate ("bicarbonate of soda"). Antacids can also be used to relieve heartburn.

- **Industrial uses**

Basic calcium hydroxide (limewater) can be used to absorb harmful acidic SO_2 gas that is released from power stations and from the burning of fossil fuels.

General experiment: Acid-base reactions

Aim:

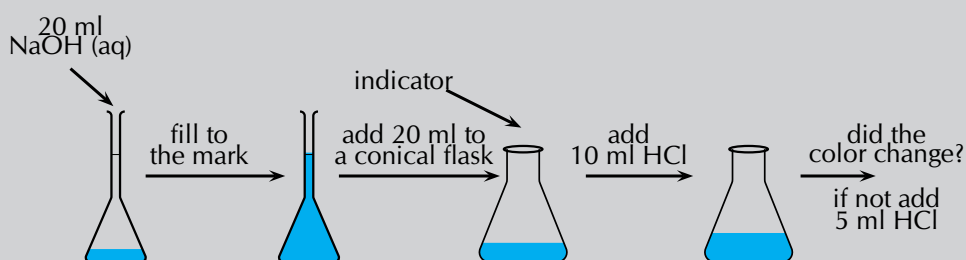
To investigate acid-base reactions.

Apparatus and materials:

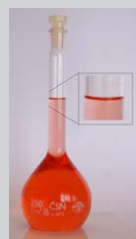
- Volumetric flask
- conical flasks

- sodium hydroxide solution
- hydrochloric acid solution
- pipette
- indicator

Method:



1. Use the pipette to add 20 ml of the sodium hydroxide solution to a volumetric flask. Fill up to the mark with water and shake well.
2. Measure 20 ml of the sodium hydroxide solution into a conical flask. Add a few drops of indicator.
3. Slowly add 10 ml of hydrochloric acid. If there is a colour change stop. If not add another 5 ml. Continue adding 5 ml increments until you notice a colour change.



Observations:

The solution changes colour after a set amount of hydrochloric acid is added.

In the above experiment you used an indicator to see when the acid had neutralised the base. Indicators are chemical compounds that change colour depending on whether they are in an acid or in a base.

Informal experiment: Indicators

Aim:

To determine which plants and foods can act as indicators.

Apparatus and materials:

- Possible indicators: Red cabbage, beetroot, berries (e.g. mulberries), curry powder, red grapes, onions, tea (rooibos or regular), baking powder, vanilla essence
- acids (e.g. vinegar, hydrochloric acid), bases (e.g. ammonia (in many household cleaners)) to test
- Beakers

Method:

1. Take a small amount of your first possible indicator (do not use the onions, vanilla essence and baking powder). Boil the substance up until the water has changed colour.
2. Filter the resulting solution into a beaker being careful not to get any plant matter into the beaker. (You can also pour the water through a colander or sieve.)
3. Pour half the resulting coloured solution into a second beaker.
4. Place one beaker onto an A4 sheet of paper labelled "acids". Place the other beaker onto a sheet of paper labelled "bases".
5. Repeat with all your other possible indicators (except the onions, vanilla essence and baking powder).
6. Into all the beakers on the acid sheet carefully pour 5 ml of acid. Record your observations.
7. Into all the beakers on the base sheet carefully pour 5 ml of base. Record your observations.

If you have more than one acid or base then you will need to repeat the above steps to get fresh indicator samples for your second acid or base. Or you can use less of the resulting coloured solution for each acid and base you want to test.
8. Observe the smell of the onions and vanilla essence. Place a small piece of onion into a beaker. This is for testing with the acid. Pour 5 ml of acid. Wave your hand over the top of the beaker to blow the air towards your nose. What do you notice about the smell of the onions? Repeat with the vanilla essence.
9. Place a small piece of onion into a beaker. This is for testing with the base. Pour 5 ml of base. Wave your hand over the top of the beaker to blow the air towards your nose. What do you notice about the smell of the onions? Repeat with the vanilla essence.
10. Finally place a teaspoon of baking powder into a beaker. Carefully pour 5 ml of acid into the beaker. Record your observations. Repeat using the base.

Observations:

Substance	Colour	Results with acid	Results with base
Red cabbage			
Beetroot			
Berries			
Curry powder			
Tea			
Red grapes			
Onions			
Vanilla essence			
Baking powder			

You should note that some of the substances change colour in the presence of either an acid or a base. The baking powder fizzes when it is in the acid solution, but no reaction is noted when it is in the base solution. Vanilla essence and onions should lose their characteristic smell when in the base.

We will now look at three specific types of acid-base reactions. In each of these types of acid-base reaction the acid remains the same but the kind of base changes. We will look at what kind of products are produced when acids react with each of these bases and what the general reaction looks like.

FACT

Vanilla and onions are known as olfactory (smell) indicators. Olfactory indicators lose their characteristic smell when mixed with acids or bases.

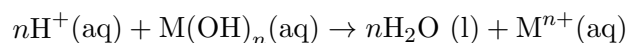
Acid and metal hydroxide

ESBQZ

When an acid reacts with a metal hydroxide a *salt and water* are formed. We have already briefly explained this. Some examples are:

- $\text{HCl (aq)} + \text{NaOH (aq)} \rightarrow \text{H}_2\text{O (l)} + \text{NaCl (aq)}$
- $2\text{HBr (aq)} + \text{Mg(OH)}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O (l)} + \text{MgBr}_2(\text{aq})$
- $3\text{HCl (aq)} + \text{Al(OH)}_3(\text{aq}) \rightarrow 3\text{H}_2\text{O (l)} + \text{AlCl}_3(\text{aq})$

We can write a general equation for this type of reaction:



Where n is the group number of the metal and M is the metal.

Exercise 13 – 3:

1. Write a balanced equation for the reaction between HNO_3 and KOH .

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1. [245V](#)



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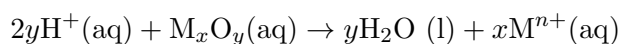
Acid and metal oxide

ESBR2

When an acid reacts with a metal oxide a *salt and water* are also formed. Some examples are:

- $2\text{HCl (aq)} + \text{Na}_2\text{O (aq)} \rightarrow \text{H}_2\text{O (l)} + 2\text{NaCl}$
- $2\text{HBr (aq)} + \text{MgO} \rightarrow \text{H}_2\text{O (l)} + \text{MgBr}_2(\text{aq})$
- $6\text{HCl (aq)} + \text{Al}_2\text{O}_3(\text{aq}) \rightarrow 3\text{H}_2\text{O (l)} + 2\text{AlCl}_3(\text{aq})$

We can write a general equation for the reaction of a metal oxide with an acid:



Where n is the group number of the metal. The x and y represent the ratio in which the metal combines with the oxide and depends on the valency of the metal.

Exercise 13 – 4:

1. Write a balanced equation for the reaction between HBr and K_2O .

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Acid and a metal carbonate

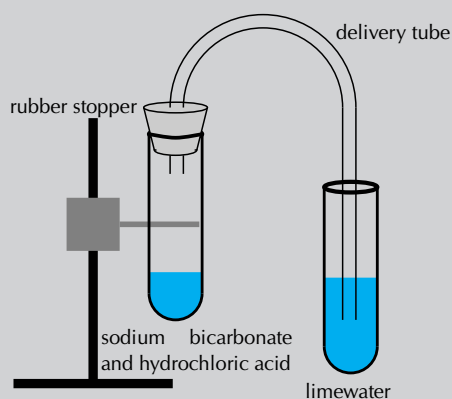
ESBR3

General experiment: The reaction of acids with carbonates

Apparatus and materials:

- Small amounts of baking powder (sodium bicarbonate)
- hydrochloric acid (dilute) and vinegar
- retort stand
- two test tubes
- one rubber stopper for the test tube
- a delivery tube
- lime water (calcium hydroxide in water)

The experiment should be set up as shown below.



Method:

1. Carefully stick the delivery tube through the rubber stopper.
2. Pour limewater into one of the test tubes.
3. Carefully pour a small amount of hydrochloric acid into the other test tube.
4. Add a small amount of sodium carbonate to the acid and seal the test tube with the rubber stopper. Place the other end of the delivery tube into the test tube containing the lime water.
5. Observe what happens to the colour of the limewater.
6. Repeat the above steps, this time using vinegar.

Observations:

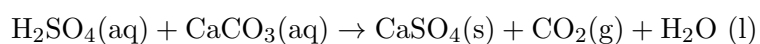
The clear lime water turns milky meaning that carbon dioxide has been produced. You may not see this for the hydrochloric acid as the reaction may happen too fast.

When an acid reacts with a metal carbonate a *salt*, *carbon dioxide* and *water* are formed. Look at the following examples:

- Nitric acid reacts with sodium carbonate to form sodium nitrate, carbon dioxide and water.



- Sulfuric acid reacts with calcium carbonate to form calcium sulfate, carbon dioxide and water.



- Hydrochloric acid reacts with calcium carbonate to form calcium chloride, carbon dioxide and water.



Exercise 13 – 5:

1. Write a balanced equation for the reaction between HCl and K_2CO_3 .

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1. [245X](#)



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Using what we have learnt about acids and bases we can now look at preparing some salts.

General experiment: Making salts

Aim:

To make some salts using acid-base reactions.

Materials:

- hydrochloric acid ($1 \text{ mol}\cdot\text{dm}^{-3}$), sulfuric acid (dilute), sodium hydroxide, copper(II) oxide, calcium carbonate
- beakers, mass meter, funnels, filter paper, bunsen burner, measuring cylinders

Method:

WARNING!

Wear gloves and safety glasses when working with sulfuric acid. Work in a well ventilated room.

Part 1

1. Measure out 20 ml of hydrochloric acid into a beaker.
2. Measure out 20 ml of sodium hydroxide and carefully add this to the beaker containing hydrochloric acid.
3. Gently heat the resulting solution until all the water has evaporated. You should have a white powder left.

Part 2

1. Carefully add 25 ml of sulfuric acid to a clean beaker.
2. Add about a small amount (about 0,5 g) of copper(II) oxide to the beaker containing sulfuric acid. Stir the solution.
3. Once all the copper(II) oxide has dissolved, add another small amount of copper(II) oxide. Repeat until no more solid dissolves and there is a small amount of undissolved solid.
4. Filter this solution and discard the filter paper.
5. Gently heat the resulting liquid. You should get a small amount of solid.

Part 3

1. Measure out 20 ml of hydrochloric acid into a new beaker.
2. Add about a small amount (about 0,5 g) of calcium carbonate to the beaker containing hydrochloric acid. Stir the solution.
3. Once all the calcium carbonate has dissolved, add another small amount of calcium carbonate. Repeat until no more solid dissolves and there is a small amount of undissolved solid.
4. Filter this solution and discard the filter paper.
5. Gently heat the resulting liquid. You should get a small amount of solid.

Observations:

In the first reaction (hydrochloric acid with sodium hydroxide) the resulting solution was clear. When this solution was heated a small amount of white powder was noted. This powder is sodium chloride.

In the second reaction (sulfuric acid with copper(II) oxide) the resulting solution was blue in colour. When this solution was heated a small amount of white powder was noted. This powder is copper sulfate.

In the third reaction (hydrochloric acid with calcium carbonate) the resulting solution was clear. When this solution was heated a small amount of white powder was noted. This powder is calcium sulfate.

Try write reaction equations for the three reactions above.

Conclusion:

We used acid-base reactions to produce different salts.

TIP

You can remember this by using OILRIG: Oxidation is Loss Reduction is Gain.

Exercise 13 – 6: Acids and bases

For each of the following reactants state what type of acid-base reaction the pair of reactants undergoes and write the balanced reaction equation.

- | | |
|--|---|
| 1. HNO_3 and $\text{Ca}(\text{OH})_2$ | 4. H_3PO_4 and KOH |
| 2. HCl and BeO | 5. HCl and MgCO_3 |
| 3. HI and K_2CO_3 | 6. HNO_3 and Al_2O_3 |

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1. [245Y](#) 2. [245Z](#) 3. [2462](#) 4. [2463](#) 5. [2464](#) 6. [2465](#)



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13.3 Redox reactions

ESBR4

If you have seen a piece of rusty metal then you have seen the end result of a redox reaction (iron and oxygen forming iron oxide). Redox reactions are also used in electrochemistry and in biological reactions.



When some reactions occur, an exchange of electrons takes place. It is this exchange of electrons that leads to the change in charge that we noted in grade 10 (chapter 18, reactions in aqueous solution). When an atom gains electrons it becomes more negative and when it loses electrons it becomes more positive.

Oxidation is the *loss* of electrons from an atom, while **reduction** is the *gain* of electrons by an atom. In a reaction these two processes occur together so that one element or compound gains electrons while the other element or compound loses electrons. This is why we call this a redox reaction. It is a short way of saying reduction-oxidation reaction!

DEFINITION: *Oxidation*

Oxidation is the *loss* of electrons by a molecule, atom or ion.

DEFINITION: *Reduction*

Reduction is the *gain* of electrons by a molecule, atom or ion.

Before we look at redox reactions we need to first learn how to tell if a reaction is a redox reaction. In grade 10 you learnt that a redox reaction involves a change in the charge on an atom. Now we will look at why this change in charge occurs.

TIP

You will notice that some elements always have the same oxidation number while other elements can change oxidation numbers depending on the compound they are in.

By giving elements an oxidation number, it is possible to keep track of whether that element is losing or gaining electrons during a chemical reaction. The loss of electrons in one part of the reaction must be balanced by a gain of electrons in another part of the reaction.

DEFINITION: *Oxidation number*

Oxidation number is the charge an atom would have if it was in a compound composed of ions.

There are a number of rules that you need to know about oxidation numbers, and these are listed below.

1. A molecule consisting of only one element always has an oxidation number of zero, since it is neutral.

For example the oxidation number of hydrogen in H_2 is 0. The oxidation number of bromine in Br_2 is also 0.

2. Monatomic ions (ions with only one element or type of atom) have an oxidation number that is equal to the charge on the ion.

For example, the chloride ion Cl^- has an oxidation number of -1 , and the magnesium ion Mg^{2+} has an oxidation number of $+2$.

3. In a molecule or compound, the sum of the oxidation numbers for each element in the molecule or compound will be zero.

For example the sum of the oxidation numbers for the elements in water will be 0.

4. In a polyatomic ion the sum of the oxidation numbers is equal to the charge.

For example the sum of the oxidation numbers for the elements in the sulfate ion (SO_4^{2-}) will be -2 .

5. An oxygen atom usually has an oxidation number of -2 . One exception is in peroxides (e.g. hydrogen peroxide) when oxygen has an oxidation number of -1 .

For example oxygen in water will have an oxidation number of -2 while in hydrogen peroxide (H_2O_2) it will have an oxidation number of -1 .

6. The oxidation number of hydrogen is often $+1$. One exception is in the metal hydrides where the oxidation number is -1 .

For example the oxidation number of the hydrogen atom in water is $+1$, while the oxidation number of hydrogen in lithium hydride (LiH) is -1 .

7. The oxidation number of fluorine is -1 .

Worked example 1: Oxidation numbers

QUESTION

Give the oxidation number of sulfur in a sulfate (SO_4^{2-}) ion

SOLUTION

Step 1: Determine the oxidation number for each atom

Oxygen will have an oxidation number of -2 . (Rule 5, this is not a peroxide.) The oxidation number of sulfur at this stage is uncertain since sulfur does not have a set oxidation number.

Step 2: Determine the oxidation number of sulfur by using the fact that the oxidation numbers of the atoms must add up to the charge on the compound

In the polyatomic SO_4^{2-} ion, the sum of the oxidation numbers must be -2 (rule 4).

Let the oxidation number of sulfur be x . We know that oxygen has an oxidation number of -2 and since there are four oxygen atoms in the sulfate ion, then the sum of the oxidation numbers of these four oxygen atoms is -8 .

Putting this together gives:

$$\begin{aligned}x + (-8) &= -2 \\x &= -2 + 8 \\&= +6\end{aligned}$$

So the oxidation number of sulfur is $+6$.

Step 3: Write down the final answer

In the sulfate ion, the oxidation number of sulfur is $+6$.

Worked example 2: Oxidation numbers

QUESTION

Give the oxidation number of both elements in ammonia (NH_3).

SOLUTION

Step 1: Determine the oxidation number for each atom

Hydrogen will have an oxidation number of $+1$ (rule 6, ammonia is not a metal hydride). At this stage we do not know the oxidation number for nitrogen.

Step 2: Determine the oxidation number of nitrogen by using the fact that the oxidation numbers of the atoms must add up to the charge on the compound

In the compound NH_3 , the sum of the oxidation numbers must be 0 (rule 3).

Let the oxidation number of nitrogen be x . We know that hydrogen has an oxidation number of +1 and since there are three hydrogen atoms in the ammonia molecule, then the sum of the oxidation numbers of these three hydrogen atoms is +3.

Putting this together gives:

$$\begin{aligned}x + (+3) &= 0 \\ &= -3\end{aligned}$$

So the oxidation number of nitrogen is -3 .

Step 3: Write the final answer

Hydrogen has an oxidation number of +1 and nitrogen has an oxidation number of -3 .

Worked example 3: Oxidation numbers

QUESTION

Give the oxidation numbers for all the atoms in sodium chloride (NaCl).

SOLUTION

Step 1: Determine the oxidation number for each atom in the compound

This is an ionic compound composed of Na^+ and Cl^- ions. Using rule 2 the oxidation number for the sodium ion is +1 and for the chlorine ion it is -1 .

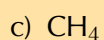
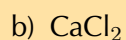
This then gives us a sum of 0 for the compound.

Step 2: Write the final answer

The oxidation numbers for sodium is +1 and for chlorine it is -1 .

Exercise 13 – 7: Oxidation numbers

1. Give the oxidation numbers for each element in the following chemical compounds:



2. Compare the oxidation numbers of:

TIP

The word species is used in chemistry to indicate either a compound, a molecule, an ion, an atom or an element.

- a) nitrogen in:
NO₂ and NO
- b) carbon in:
CO₂ and CO

- c) chromium in:
Cr₂O₇²⁻ and CrO₄⁻
- d) oxygen in:
H₂O and H₂O₂

- e) hydrogen in:
NaH and H₂O

3. Give the oxidation numbers for each of the elements in all the compounds. State if there is any difference between the oxidation number of the element in the reactant and the element in the product.

- a) C (s) + O₂(g) → CO₂(g)
- b) N₂(g) + 3H₂(g) → 2NH₃(g)

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1a. [2466](#) 1b. [2467](#) 1c. [2468](#) 1d. [2469](#) 2a. [246B](#) 2b. [246C](#)
2c. [246D](#) 2d. [246F](#) 2e. [246G](#) 3a. [246H](#) 3b. [246J](#)



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Redox reactions

ESBR6

Now that we know how to determine the oxidation number of a compound, we will go on to look at how to use this knowledge in reactions.

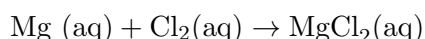
Oxidation and reduction

By looking at how the oxidation number of an element changes during a reaction, we can easily see whether that element is being **oxidised** (lost electrons) or **reduced** (gained electrons).

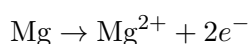
If the oxidation number of a species becomes more positive, the species has been **oxidised** and if the oxidation number of a species becomes more negative, the species has been **reduced**.

We will use the reaction between magnesium and chlorine as an example.

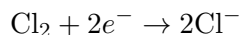
The chemical equation for this reaction is:



As a *reactant*, magnesium has an oxidation number of zero, but as part of the *product* magnesium chloride, the element has an oxidation number of +2. Magnesium has *lost* two electrons and has therefore been *oxidised* (note how the oxidation number becomes more positive). This can be written as a **half-reaction**. The half-reaction for this change is:



As a *reactant*, chlorine has an oxidation number of zero, but as part of the *product* magnesium chloride, the element has an oxidation number of -1 . Each chlorine atom has *gained* an electron and the element has therefore been *reduced* (note how the oxidation number becomes more negative). The half-reaction for this change is:



DEFINITION: *Half-reaction*

A half reaction is either the oxidation or reduction reaction part of a redox reaction.

In the two half-reactions for a redox reaction the number of electrons donated is exactly the same as the number of electrons accepted. We will use this to help us balance redox reactions.

Two further terms that we use in redox reactions and that you may see are reducing agents and oxidising agents.

An element that is **oxidised** is called a **reducing agent**, while an element that is **reduced** is called an **oxidising agent**.

You can remember this by thinking of the fact that when a compound is oxidised, it causes another compound to be reduced (the electrons have to go somewhere and they go to the compound being reduced).

Redox reactions

DEFINITION: *Redox reaction*

A redox reaction is one involving oxidation and reduction, where there is always a change in the oxidation numbers of the elements involved. Redox reactions involve the transfer of electrons from one compound to another.

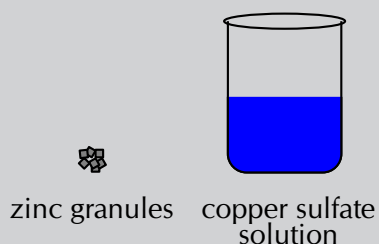
Informal experiment: Redox reaction - displacement reaction

Aim:

To investigate the redox reaction between copper sulfate and zinc.

Materials:

- A few granules of zinc
- 15 ml copper (II) sulfate solution (blue colour)
- glass beaker



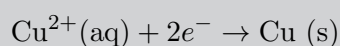
Method:

Add the zinc granules to the copper sulfate solution and observe what happens. What happens to the zinc granules? What happens to the colour of the solution?

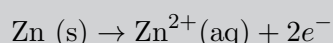
Results:

- Zinc becomes covered in a layer that looks like copper.
- The blue copper sulfate solution becomes clearer.

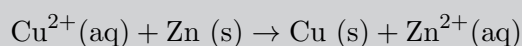
Cu^{2+} ions from the CuSO_4 solution are **reduced** to form copper metal. This is what you saw on the zinc crystals. The reduction of the copper ions (in other words, their removal from the copper sulfate solution), also explains the change in colour of the solution (copper ions in solution are blue). The equation for this reaction is:



Zinc is **oxidised** to form Zn^{2+} ions which are clear in the solution. The equation for this reaction is:



The overall reaction is:

**Conclusion:**

A redox reaction has taken place. Cu^{2+} ions are reduced and the zinc is oxidised. This is a displacement reaction as the zinc replace the copper ions to form zinc sulfate.

Informal experiment: Redox reaction - synthesis reaction**Aim:**

To investigate the redox reaction that occurs when magnesium is burnt in air.

Materials:

A strip of magnesium; bunsen burner; tongs; glass beaker.

Method:

WARNING!

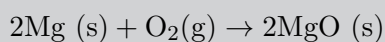
Do not look directly at the flame.

1. Light the bunsen burner and use a pair of tongs to hold the magnesium ribbon in the flame.
2. Hold the lit piece of magnesium over a beaker. What do you observe?

Results:

The magnesium burns with a bright white flame. When the magnesium is held over a beaker, a fine powder is observed in the beaker. This is magnesium oxide.

The overall reaction is:



Conclusion:

A redox reaction has taken place. Magnesium is oxidised and the oxygen is reduced. This is a synthesis reaction as we have made magnesium oxide from magnesium and oxygen.

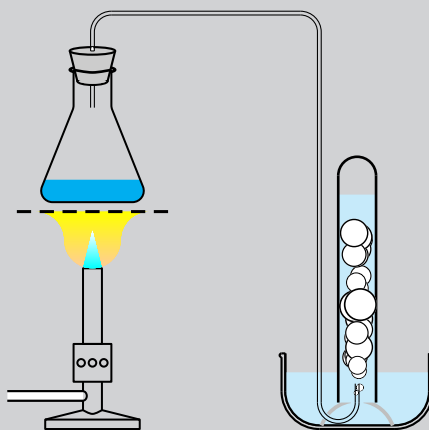
Informal experiment: Redox reaction - decomposition reaction

Aim:

To investigate the decomposition of hydrogen peroxide.

Materials:

Dilute hydrogen peroxide (about 3%); manganese dioxide; test tubes; a water bowl; stopper and delivery tube, Bunsen burner



WARNING!

Hydrogen peroxide can cause chemical burns. Work carefully with it.

Method:

1. Put a small amount (about 5 ml) of hydrogen peroxide in a test tube.
2. Set up the apparatus as shown above.
3. Very carefully add a small amount (about 0,5 g) of manganese dioxide to the test tube containing hydrogen peroxide.

Results:

You should observe a gas bubbling up into the second test tube. This reaction happens quite rapidly.

The overall reaction is:



Conclusion:

A redox reaction has taken place. H_2O_2 is both oxidised and reduced in this decomposition reaction.

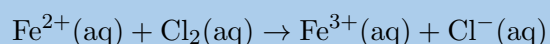
► See video: 246K at www.everythingscience.co.za

Using what you have learnt about oxidation numbers and redox reactions we can balance redox reactions in the same way that you have learnt to balance other reactions. The following worked examples will show you how.

Worked example 4: Balancing redox reactions

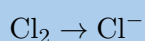
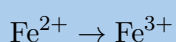
QUESTION

Balance the following redox reaction:



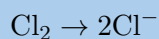
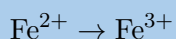
SOLUTION

Step 1: Write a reaction for each compound



Step 2: Balance the atoms on either side of the arrow

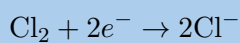
We check that the atoms on both sides of the arrow are balanced:



Step 3: Add electrons to balance the charges

We now add electrons to each reaction so that the charges balance.

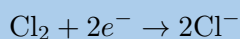
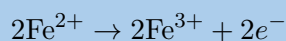
We add the electrons to the side with the greater positive charge.



Step 4: Balance the number of electrons

We now make sure that the number of electrons in both reactions is the same.

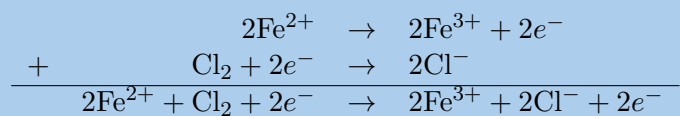
The reaction for iron has one electron, while the reaction for chlorine has two electrons. So we must multiply the reaction for iron by 2 to ensure that the charges balance.



We now have the two half-reactions for this redox reaction.

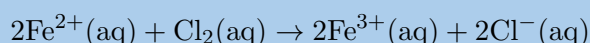
The reaction for iron is the oxidation half-reaction as iron became more positive (lost electrons). The reaction for chlorine is the reduction half-reaction as chlorine has become more negative (gained electrons).

Step 5: Combine the two half-reactions



Step 6: Write the final answer

Cancelling out the electrons gives:



Note that we leave the co-efficients in front of the iron ions since the charge on the left hand side has to be the same as the charge on the right hand side in a redox reaction.

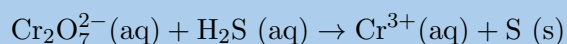
In the example above we did not need to know if the reaction was taking place in an acidic or basic medium (solution). However if there is hydrogen or oxygen in the reactants and not in the products (or if there is hydrogen or oxygen in the products but not in the reactants) then we need to know what medium the reaction is taking place in. This will help us to balance the redox reaction.

If a redox reaction takes place in an **acidic** medium then we can add water molecules to either side of the reaction equation to balance the number of oxygen atoms. We can also add hydrogen ions to balance the number of hydrogen atoms. We do this because we are writing the net ionic equation (showing only the ions involved and often only the ions containing the elements that change oxidation number) for redox reactions and not the net reaction equation (showing all the compounds that are involved in the reaction). If a Bronsted acid is dissolved in water then there will be free hydrogen ions.

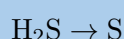
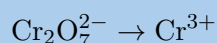
If a redox reaction takes place in an **basic** medium then we can add water molecules to either side of the reaction equation to balance the number of oxygen atoms. We can also add hydroxide ions (OH^{-}) to balance the number of hydrogen atoms. We do this because we are writing the net ionic equation (showing only the ions involved and often only the ions containing the elements that change oxidation number) for redox reactions and not the net reaction equation (showing all the compounds that are involved in the reaction). If a Bronsted base is dissolved in water then there will be free hydroxide ions.

QUESTION

Balance the following redox reaction:



The reaction takes place in an acidic medium.

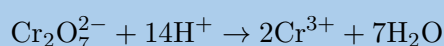
SOLUTION
Step 1: Write a reaction for each compound

Step 2: Balance the atoms on either side of the arrow

We check that the atoms on both sides of the arrow are balanced.

In the first reaction we have 2 chromium atoms and 7 oxygen atoms on the left hand side. On the right hand side we have 1 chromium atom and no oxygen atoms. Since we are in an acidic medium we can add water to the right hand side to balance the number of oxygen atoms. We also multiply the chromium by 2 on the right hand side to make the number of chromium atoms balance.

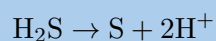


Now we have hydrogen atoms on the right hand side, but not on the left hand side so we must add 14 hydrogen ions to the left hand side (we can do this because the reaction is in an acidic medium):



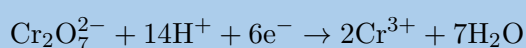
We do not use water to balance the hydrogens as this will make the number of oxygen atoms unbalanced.

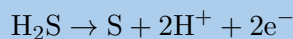
For the second part of the reaction we need to add 2 hydrogen ions to the right hand side to balance the number of hydrogens:


Step 3: Add electrons to balance the charges

We now add electrons to each reaction so that the charges balance.

We add the electrons to the side with the greater positive charge.

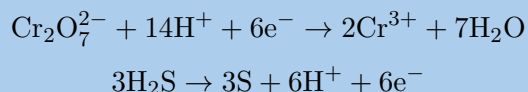




Step 4: Balance the number of electrons

We now make sure that the number of electrons in both reactions is the same.

The reaction for chromium has 6 electrons, while the reaction for sulfur has 2 electrons. So we must multiply the reaction for sulfur by 3 to ensure that the charges balances.

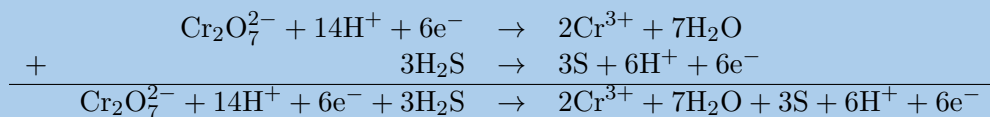


We now have the two half-reactions for this redox reaction.

The reaction involving sulfur is the oxidation half-reaction as sulfur became more positive (lost electrons). The reaction for chromium is the reduction half-reaction as chromium has become more negative (gained electrons).

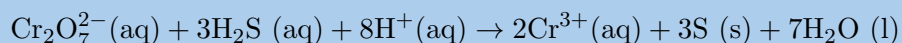
Step 5: Combine the two half-reactions

We combine the two half-reactions:



Step 6: Write the final answer

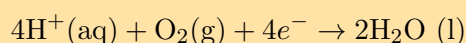
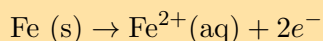
Crossing off the electrons and hydrogen ions gives:



In Grade 12, you will go on to look at electrochemical reactions, and the role that electron transfer plays in this type of reaction.

Exercise 13 – 8: Redox reactions

1. Consider the following chemical equations:



Which one of the following statements is correct?

- a) Fe is oxidised and H^+ is reduced
- b) Fe is reduced and O_2 is oxidised
- c) Fe is oxidised and O_2 is reduced
- d) Fe is reduced and H^+ is oxidised

(DoE Grade 11 Paper 2, 2007)

2. Which one of the following reactions is a redox reaction?

- a) $\text{HCl (aq)} + \text{NaOH (aq)} \rightarrow \text{NaCl (aq)} + \text{H}_2\text{O (l)}$
- b) $\text{AgNO}_3(\text{aq}) + \text{NaI (aq)} \rightarrow \text{AgI (s)} + \text{NaNO}_3(\text{aq})$
- c) $2\text{FeCl}_3(\text{aq}) + 2\text{H}_2\text{O (l)} + \text{SO}_2(\text{aq}) \rightarrow \text{H}_2\text{SO}_4(\text{aq}) + 2\text{HCl (aq)} + 2\text{FeCl}_2(\text{aq})$
- d) $\text{BaCl}_2(\text{aq}) + \text{MgSO}_4(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{BaSO}_4(\text{s})$

3. Balance the following redox reactions:

- a) $\text{Zn (s)} + \text{Ag}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Ag (s)}$
- b) $\text{Cu}^{2+}(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{Cu (s)} + \text{Cl}_2(\text{g})$
- c) $\text{Pb}^{2+}(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{Pb (s)} + \text{Br}_2(\text{aq})$
- d) $\text{HCl (aq)} + \text{MnO}_2(\text{s}) \rightarrow \text{Cl}_2(\text{g}) + \text{Mn}^{2+}(\text{aq})$
This reaction takes place in an acidic medium.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. [246M](#) 2. [246N](#) 3a. [246P](#) 3b. [246Q](#) 3c. [246R](#) 3d. [246S](#)



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13.4 Chapter summary

ESBR7

▶ See presentation: [246T](#) at www.everythingscience.co.za

- There are many different **types of reactions** that can take place. These include acid-base and redox reactions.
- The **Arrhenius** definition of acids and bases defines an acid as a substance that increases the concentration of hydronium ions (H_3O^+) in a solution. A base is a substance that increases the concentration of hydroxide ions (OH^-) in a solution. However this description only applies to substances that are in water.
- The **Bronsted-Lowry** model defines acids and bases in terms of their ability to donate or accept protons.
- A Bronsted-Lowry **acid** is a substance that gives away protons (hydrogen cations H^+), and is therefore called a **proton donor**.

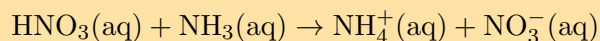
- A Bronsted-Lowry **base** is a substance that takes up protons (hydrogen cations H^+), and is therefore called a **proton acceptor**.
- An **amphoteric** substance is one that can react as either an acid or base. Water (H_2O) is an example of an amphoteric substance.
- An **amphiprotic** is one that can react as either a proton donor (Bronsted-Lowry acid) or as a proton acceptor (Bronsted-Lowry base). HCO_3^- and HSO_4^- are examples of amphiprotic substances.
- A **conjugate acid-base pair** refers to two compounds in a reaction (one reactant and one product) that transform or change into the other through the loss or gain of a proton.
- The reaction between an acid and a base is a **neutralisation** reaction.
- Acids and bases are used in domestic uses (for example calcium carbonate on acidic soil) in biology (for example in antacids for stomach ulcers) and in industry (for example in absorbing harmful SO_2 gas),.
- Indicators are chemical compounds that change colour depending on whether they are in an acid or in a base.
- When an acid reacts with a metal hydroxide a **salt** and water are formed. The salt is made up of a cation from the base and an anion from the acid. An example of a salt is potassium chloride (KCl), which is the product of the reaction between potassium hydroxide (KOH) and hydrochloric acid (HCl).
- When an acid reacts with a metal oxide a salt and water are formed. An example is the reaction between magnesium oxide (MgO) and hydrochloric acid (HCl).
- When an acid reacts with a metal carbonate a salt, water and carbon dioxide are formed. An example is the reaction between calcium carbonate (CaCO_3) and hydrochloric acid (HCl).
- **Oxidation** is the loss of electrons by a molecule, atom or ion.
- **Reduction** is the gain of electrons by a molecule, atom or ion.
- A **redox reaction** is one involving oxidation and reduction, where there is always a change in the oxidation numbers of the elements involved. Redox reactions involve the transfer of electrons from one compound to another.
- An **oxidation number** is the charge an atom would have if it was in a compound composed of ions.
- If the oxidation number of a species becomes more positive, the species has been oxidised and if the oxidation number of a species becomes more negative, the species has been reduced.
- A half reaction is either the oxidation or reduction reaction part of a redox reaction. In the two half-reactions for a redox reaction the number of electrons donated is exactly the same as the number of electrons accepted.
- An element that is **oxidised** is called a **reducing agent**, while an element that is **reduced** is called an **oxidising agent**.

Exercise 13 – 9:

1. Give one word/term for each of the following descriptions:

- a) A chemical reaction during which electrons are transferred.
- b) A substance that takes up protons and is said to be a proton acceptor.
- c) The loss of electrons by a molecule, atom or ion.
- d) A substance that can act as either an acid or as a base.

2. Given the following reaction:

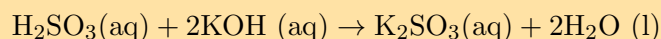


Which of the following statements is true?

- a) HNO_3 accepts protons and is a Bronsted-Lowry base
 - b) HNO_3 accepts protons and is a Bronsted-Lowry acid
 - c) NH_3 donates protons and is a Bronsted-Lowry acid
 - d) HNO_3 donates protons and is a Bronsted-Lowry acid
3. When chlorine water (Cl_2 dissolved in water) is added to a solution of potassium bromide, bromine is produced. Which one of the following statements concerning this reaction is correct?
- a) Br is oxidised
 - b) Cl_2 is oxidised
 - c) Br^- is the oxidising agent
 - d) Cl^- is the oxidising agent

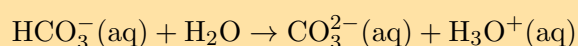
(IEB Paper 2, 2005)

4. Given the following reaction:

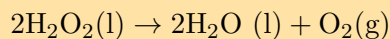


Which substance acts as the acid and which substance acts as the base?

5. Use balanced chemical equations to explain why HSO_4^- is amphoteric.
6. Milk of magnesia is an example of an antacid and is only slightly soluble in water. Milk of magnesia has the chemical formula $\text{Mg}(\text{OH})_2$ and is taken as a powder dissolved in water. Write a balanced chemical equation to show how the antacid reacts with hydrochloric acid (the main acid found in the stomach).
7. In an experiment sodium carbonate was used to neutralise a solution of hydrochloric acid. Write a balanced chemical equation for this reaction.
8. Write a balanced chemical equation for phosphoric acid (H_3PO_4) reacting with calcium oxide (CaO).
9. Label the acid-base conjugate pairs in the following equation:



10. Look at the following reaction:



a) What is the oxidation number of the oxygen atom in each of the following compounds?

- i. H_2O_2
- ii. H_2O
- iii. O_2

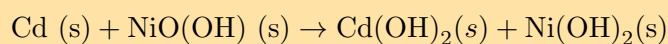
b) Does the hydrogen peroxide (H_2O_2) act as an oxidising agent or a reducing agent or both, in the above reaction? Give a reason for your answer.

11. Challenge question: Zinc reacts with aqueous copper sulfate ($\text{CuSO}_4(\text{aq})$) to form zinc sulfate ($\text{ZnSO}_4(\text{aq})$) and copper ions. What is the oxidation number for each element in the reaction?

12. Balance the following redox reactions:

- a) $\text{Pb}(\text{s}) + \text{Ag}^+(\text{aq}) \rightarrow \text{Pb}^{2+}(\text{aq}) + \text{Ag}(\text{s})$
- b) $\text{Zn}^{2+}(\text{aq}) + \text{I}^-(\text{aq}) \rightarrow \text{Zn}(\text{s}) + \text{I}_2(\text{g})$
- c) $\text{Fe}^{3+}(\text{aq}) + \text{NO}_2(\text{aq}) \rightarrow \text{Fe}(\text{s}) + \text{NO}_3^-(\text{aq})$
This reaction takes place in an acidic medium.

13. A nickel-cadmium battery is used in various portable devices such as calculators. This battery uses a redox reaction to work. The equation for the reaction is:



This reaction takes place in a basic medium. Balance the equation.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

- | | | | | | |
|----------|----------|-----------|-----------|-----------|----------|
| 1a. 246V | 1b. 246W | 1c. 246X | 1d. 246Y | 2. 246Z | 3. 2472 |
| 4. 2473 | 5. 2474 | 6. 2475 | 7. 2476 | 8. 2477 | 9. 2478 |
| 10. 2479 | 11. 247F | 12a. 247G | 12b. 247H | 12c. 247J | 13. 247K |



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The lithosphere

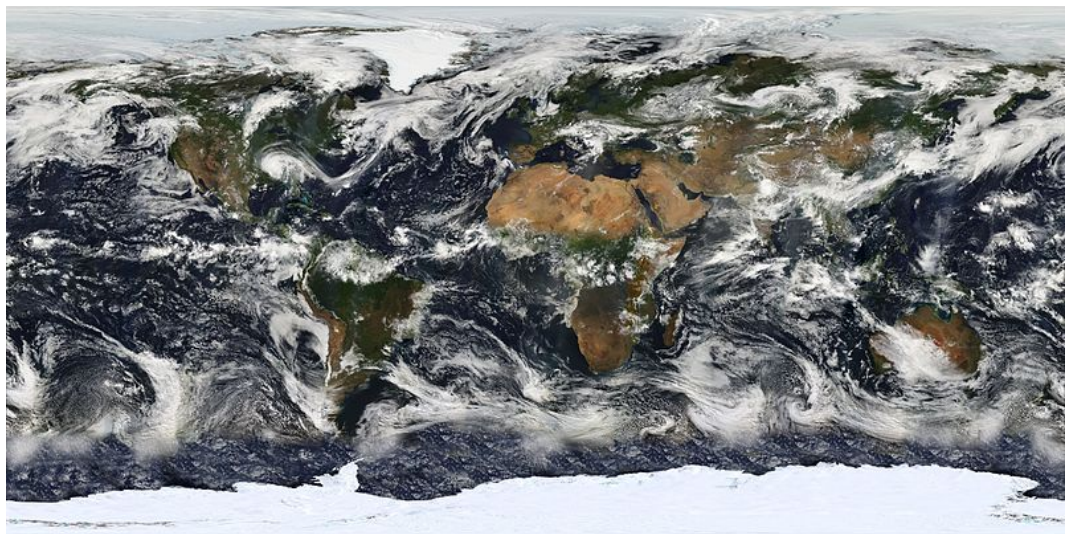
14.1	<i>Introduction</i>	470
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14.1 Introduction

ESBR8

In Grade 10 we studied the elements. We learnt about the different elements and the compounds that could be formed from those elements. But where do all these elements come from? Where did mankind find them and how has he used them?

This chapter will explore the part of the Earth known as the lithosphere. We will find out what the lithosphere is and how the elements are distributed within it.



14.2 The lithosphere

ESBR9

If we were to cut the Earth in half we would see that our planet is made up of a number of layers, namely the **core** at the centre (separated into the inner and outer core), the **mantle**, the **upper mantle**, the **crust** and the **atmosphere** (Figure 14.1). The core is made up mostly of iron. The mantle, which lies between the core and the crust, consists of molten rock, called **magma** which moves continuously because of convection currents. The crust is the thin, hard outer layer that “floats” on the magma of the mantle. It is the upper part of the mantle and the crust that make up the **lithosphere** (lith means types of stone and sphere refers to the round shape of the Earth).

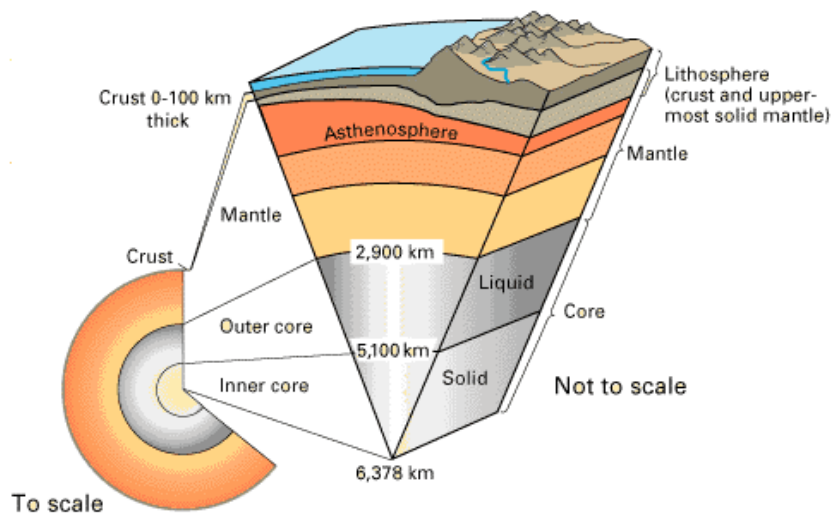


Figure 14.1: A cross-section through the Earth to show its different layers.

DEFINITION: *Lithosphere*

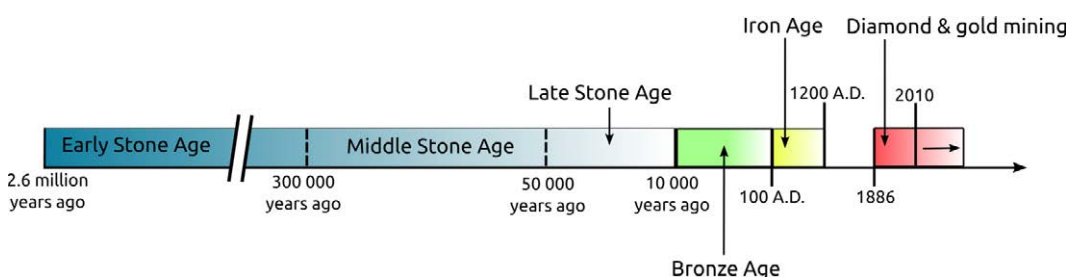
The lithosphere is the solid outermost shell of our planet. The lithosphere includes the crust and the upper part of the mantle, and is made up of material from both the continents and the oceans on the Earth's surface.

The lithosphere is vital to humans as this is the part of the Earth that we live on and can easily access. This is the part of the Earth that supports us, provides for us and gives us a wealth of materials to use. We are now going to explore how man slowly became aware of the wealth of minerals at his feet and how he has learnt to use these minerals for his benefit.

History of mankind

ESBRB

The ancient history of humankind can be divided into several periods: the Stone Age or Palaeolithic period (from approximately 2,6 million years to 10 000 years ago); the Bronze Age or Mesolithic period (from approximately 10 000 B.C. to 100 A.D.); and the Iron Age or Neolithic period (approximately 100 A.D. to 1200 A.D.) In many places in the world these periods overlapped in time and it is important to understand that the above dates are very general approximations.



The Stone Age can be further divided into three periods: the Early Stone Age (2,6 million – 300 000 years ago); the Middle Stone Age (approximately 300 000 – 50 000 years ago); and the Late Stone Age or (approximately 50 000 – 10 000 years ago).

FACT

The largest Iron Age site in South Africa is found at Mapungubwe, in the Limpopo Province. Mapungubwe (inhabited from about 1000 – 1300 A.D.) is thought to be the ancient capital of an African king. The site was rediscovered in 1932 and has proved to be a treasure trove of archaeological evidence for an advanced society. The remains of royal burial sites, houses, tools, art and farming implements have been found. Gold-plated jewellery, sculptures and artifacts (thought to have belonged to members of the royal family) have also been discovered at the site.

Age	Time	Information	Major material	Images
Early stone age	2,6 million - 300 000 years ago	Basic stone tools were used. Mankind mainly used the rocks and stones he found lying around him. Evidence for this has been found in the Sterkfontein caves in the Cradle of Humankind.	Stone	
Middle stone age	300 000 - 50 000 years ago	Mankind learnt to use fire to treat stones before they were made into tools. Stone tools became more refined during this period. Evidence for this has been found at Swartkrans (Cradle of Humankind), Montagu Cave (Klein Karoo), Klasies River Mouth (Tsitsikamma), Stilbaai and Blombos Cave (Southern Cape).	Stone treated with fire	
Late stone age	50 000 - 10 000 years ago	Mankind started turning to other materials that were easily available. He started working with different materials and seeing how these new materials could be combined with stone. Evidence for this has been found in the Melkhoutboom Cave, in the Surberg Mountains (Eastern Cape).	Stone with other natural materials	
Bronze age	10 000 B.C. - 100 A.D.	The Bronze Age saw the first use of smelted copper and bronze in the manufacture of tools and weapons. Evidence for this has been found in North Africa.	Copper and tin	
Iron age	100 A.D. - 1200 A.D.	Primitive mining for materials occurred during this period. Furnaces were used to heat metal, in order to smelt it into weapons and tools. Evidence for this has been found at Melville Koppies and at Mapungubwe (Limpopo).	Iron and other metals	

Table 14.1: The different ages of civilisation.

After the Iron Age ended, mankind started to explore many more ways to get at the precious metals that they were using. Mining for the minerals started to become more common and man began to look for new sources of the metals and minerals he needed.

Many different cultures in Africa used gold, diamonds and other precious metals for different things. A large number of these artefact's were looted from these people and taken to Europe by the colonists and explorers that came from Europe. Efforts have now begun to return these stolen artifacts to the people they came from, but many museums still hold onto their collections.

When colonists and explorers first began exploring and settling in Africa they started looking at ways to use the natural resources they found.

Although diamonds had long been known to the indigenous people of South Africa, colonists found them on a farm outside Kimberley (Northern Cape) in the 1860's. The original hill on which the diamonds were found was mined extensively by the hands of thousands of hopeful diamond prospectors who rushed to the area. Today the site of that hill is now the Big Hole in Kimberley – one of the largest holes in the world that was dug by hand. In 1888, several small diamond mining companies merged to form De Beers, the company that still dominates the global diamond mining and retail industry today.

Around the same time (1880), the colonists discovered gold reefs just south of present-day Johannesburg. The rush of gold-prospectors to the area resulted in the birth of the city of Johannesburg. A massive labour force was required to dig deep into the bedrock to extract the rock containing the gold. Indeed, huge numbers of mine workers are still employed by gold mines in South Africa today, but (as with diamond mining too), due to technological advancements over the last 150 years, much of their work is now aided by industrial mining equipment.

Many farmers and indigenous people were forced to work in the new diamond and gold mines. These people were taken by force from their homes and put to work on the mines. Many of them died from a result of the poor working conditions and from mining disasters that occurred.

FACT

The deepest mine in the world is currently the TauTona Mine (or Western Deep No. 3 Shaft) outside Carletonville, Gauteng. Owned by AngloGold Ashanti, it is 3,9 km deep and has about 800 km of tunnels. The rock face at that depth reaches temperatures of 60°C and air-conditioners have to be used to cool the air in the mine from 55°C to 28°C.

Group discussion: Minerals and the rights to own them

Many indigenous people share the same central belief that the land and all it produces are for all the people to use equally. The land and what it produces is often vital to the survival of these people.

When the colonists came to Southern Africa, they largely ignored the indigenous people and exploited them for the knowledge they held and the work they could provide. A great number of atrocities were committed against the indigenous people.

Now the indigenous people are rising up and asking that what is rightfully theirs should be returned to them. Organisations (such as the UN working group on indigenous people) have formed to address these issues.

Shortly after De Beers formed they claimed the right to all the diamonds in the area and anyone found with diamonds could be killed for taking the diamonds illegally. Other large mining companies have tried to claim the rights to the minerals that they produce.

In groups or as a class discuss whether or not a few select people should hold the rights to the land and the minerals in it. Who owns the minerals? Should big corporates hold the rights? Or should it belong to all the people?

Now that we know what mankind used to make tools and shelter from, we will take a look at why he chose the materials he did. In the early ages (early and middle stone age) mankind used whatever was to hand and easy to get to. Later on (from the late stone age to the present day) he started wondering how these different materials could be improved. So what exactly is in the lithosphere and how did mankind use this knowledge?

The crust is made up of about 80 elements, which occur in over 2000 different compounds and minerals. However, most of the mass of the material in the crust is made up of only 8 of these elements. These are oxygen (O), silicon (Si), aluminium (Al), iron (Fe), calcium (Ca), sodium (Na), potassium (K) and magnesium (Mg). These elements are seldom found in their pure form, but are usually part of other more complex **minerals**. A mineral is a compound that is formed through geological processes, which give it a particular structure. A mineral could be a pure element, but more often minerals are made up of many different elements combined. *Quartz* is just one example. It is a mineral that is made up of silicon and oxygen. Some more examples are shown in Table 14.2.

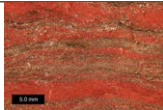
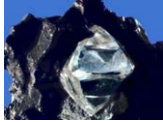
Element	Most common mineral	Chemistry	
Gold	Calaverite or pure element	Au (pure element) or AuTe ₂ (Calaverite, a gold mineral)	
Iron	Hematite	Fe ₂ O ₃ (iron oxide)	
Copper	Pure element or chalcocite	Cu (pure element) or Cu ₂ S (copper sulfide)	
Carbon	Diamond, graphite, coal	C (pure element)	
Platinum	Pure element, combined with other elements	Pt (pure element)	
Zinc	Sphalerite	ZnS	
Manganese	Manganese dioxide	MnO ₂	
Chromium (chrome)	Chromite	FeCr ₂ O ₄	

Table 14.2: Table showing examples of minerals and their chemistry.

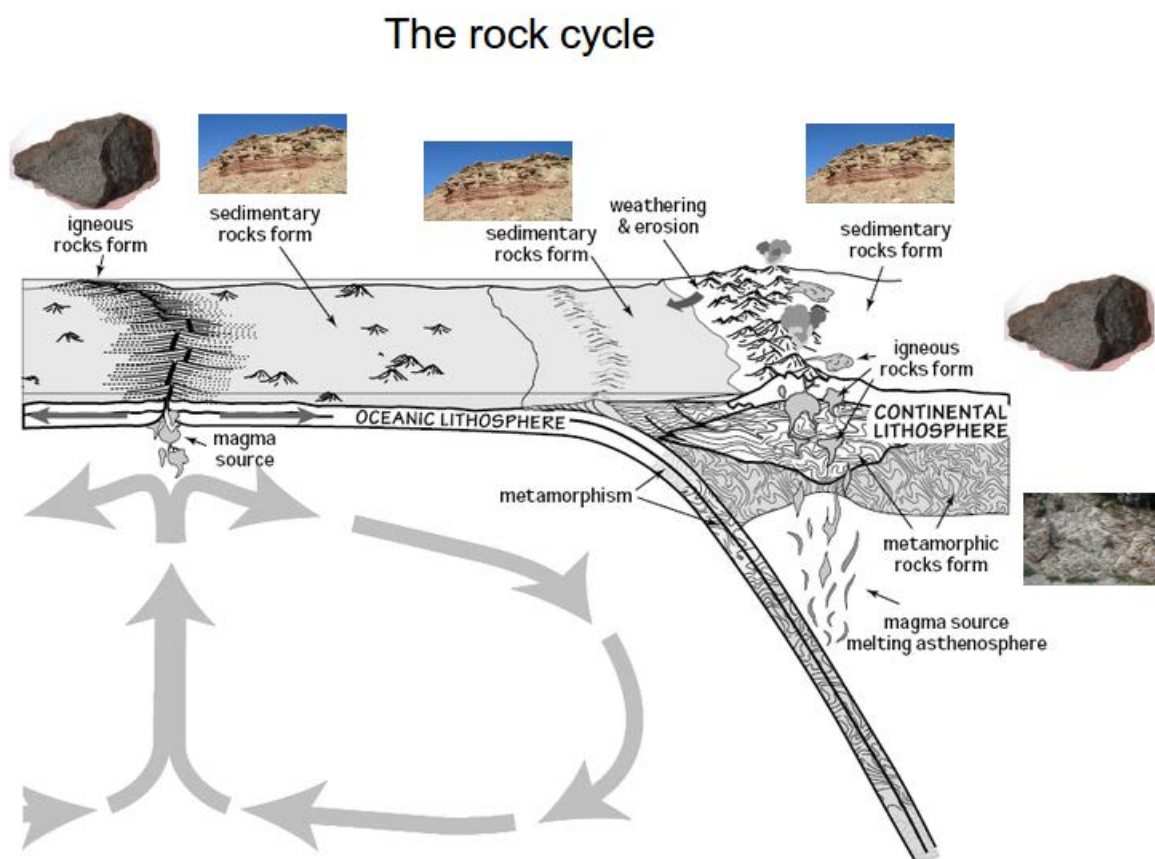
DEFINITION: Mineral

Minerals are natural compounds formed through geological processes. The term “mineral” includes both the material’s chemical composition and its structure. Minerals range in composition from pure elements to complex compounds.

In this chapter we will mainly look at metal minerals (gold, copper, iron). There are also non-metal minerals (sand, stone) and fuel minerals (coal, oil).

A **rock** is a combination of one or more minerals. *Granite* for example, is a rock that is made up of minerals such as SiO_2 , Al_2O_3 , CaO , K_2O , Na_2O and others. There are three different types of rocks: **igneous**, **sedimentary** and **metamorphic**. Igneous rocks (e.g. granite, basalt) are formed when magma is brought to the Earth’s surface as lava, and then solidifies. Sedimentary rocks (e.g. sandstone, limestone) form when rock fragments, organic matter or other sediment particles are deposited and then compacted over time until they solidify. Metamorphic rock is formed when any other rock types are subjected to intense heat and pressure over a period of time. Examples include slate and marble.

The figure below shows how these different types of rock are formed in the lithosphere.



Many of the elements that are of interest to us (e.g. gold, iron, copper), are unevenly distributed in the lithosphere. In places where these elements are abundant, it is profitable to extract them (e.g. through mining) for economic purposes. If their concentration is very low, then the cost of extraction becomes more than the money that would be made if they were sold. Rocks that contain valuable minerals are called **ores**.

FACT

A **gemstone** (also sometimes called a **gem**, **semi-precious stone** or **precious stone**), is a highly attractive and valuable piece of mineral which, when cut and polished, is used in jewellery and other adornments. Examples of gemstones are amethyst, diamond, cat's eye and sapphire.

As humans, we are particularly interested in the ores that contain metal elements, and also in those minerals that can be used to produce energy.

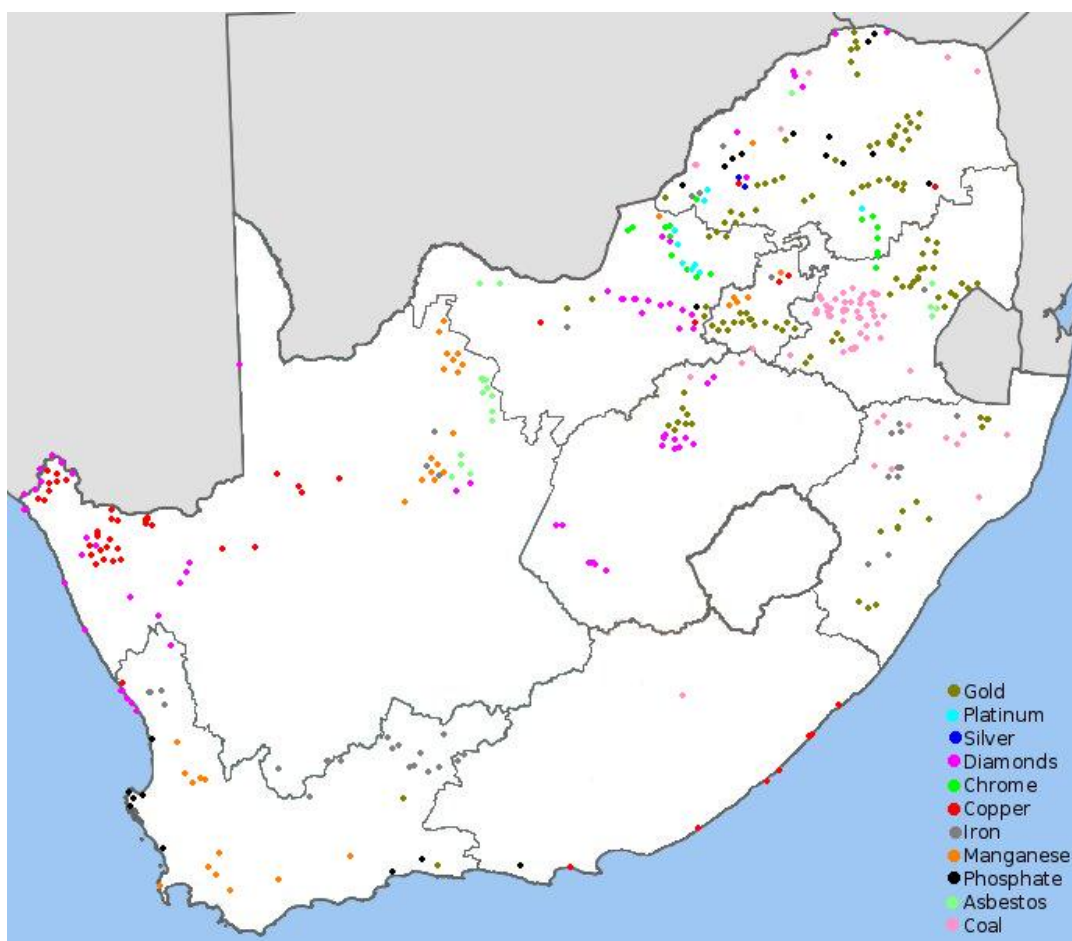


Figure 14.2: Location of minerals in South Africa.

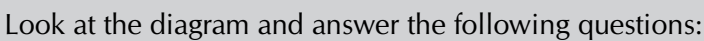
DEFINITION: *Ore*

An ore is a volume of rock that contains minerals which make it valuable for mining.

► See video: [247M](#) at www.everythingscience.co.za

Activity:

The following diagram shows the abundance of all the elements.



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FACT

Indigenous people often have unique ways of finding mineral deposits, such as observing the behaviour of animals or noting the abundance of particular plants.

FACT

Coal miners used to take a canary with them down the mines. If the canary died, they knew that there was a high level of gas building up and that it was not safe to stay down in the mine.

improved and the amount of mineral that could be extracted increased greatly. Large tunnels started to be cut into the Earth to access minerals buried deep underground. New processing methods meant new kinds of metals could be mined.

The main stages in mining are: exploration, extraction, refining, manufacturing and marketing. We will take a brief look at exploration, extraction and refining.

Exploration

ESBRF

Ancient humans did not worry about whether they could make money from the ores they mined. All they cared about was accessing metals and minerals that would help them eat, keep warm and beat the neighbouring tribes.

In modern times money matters and so the first step in mining is to find a suitably sized deposit of ore. For example, diamonds were found in Kimberley and gold ore was found in Johannesburg. These deposits were large enough to make mining for the gold and diamonds profitable.

As the number and size of known deposits is shrinking, geologists (who study the lithosphere) are developing new ways of finding suitable deposits. Geologists will often spend years learning how to find ores and then spend years exploring for the ores.

Obtaining the ore

ESBRG

Once a suitably sized deposit of a mineral has been found, mining can begin. In ancient times mining and mineral processing were very primitive and the main method used to extract ores was digging by hand.

Mining is largely divided into surface and underground mining. Minerals are often found in river beds, beach sands and other sandy areas. These are known as alluvial deposits. These minerals can be fairly easily removed by surface mining techniques. Other minerals occur in long streaks known as veins or in pipes, and underground mining techniques are used to access these minerals.

Surface mining includes open pit mines, quarrying, strip mining and landfill mining. Coal and copper are often mined in this way.

Underground mining mainly consists of digging tunnels and shafts into the Earth's crust. Gold is often mined in this way. Underground mining is more risky than surface mining as tunnels can collapse and dangerous gases build up underground.

We will look more at underground and surface mining shortly.

Extracting the minerals

ESBRH

Once the ore has been mined it is usually crushed into smaller pieces that can be processed more easily. The mineral then needs to be removed from the ore.

Ancient peoples soon learnt that fire could be used to refine and purify the metals. This technique is known as smelting. This was the earliest method used to extract the

mineral from the ore.

Smelting involves heating the ore to a very high temperature, often in a blast furnace. A reducing agent is usually added and the mineral is removed using chemical reactions.

In modern times several new techniques are used to extract the mineral from the ore, in addition to smelting. The most common methods are flotation, leaching and use of redox reactions.

Leaching involves mixing the ore with a carefully chosen liquid that dissolves either the mineral or the unwanted minerals. The liquid is often an acid.

Flotation involves the use of air bubbles to separate the valuable minerals from the unwanted rock. The valuable mineral becomes attached to the air bubbles and rises to the top of the mixture from where they can be removed.

The following table lists several metals and the typical methods used to mine and extract them.

Metal	Mining techniques	Extraction techniques
Gold	Underground mining (shaft mining) and surface mining.	Gold cyanidation is used. The ore is chemically treated to extract the gold.
Iron	Surface mining (open pit mining).	Smelting and chemical reduction.
Phosphate	Surface mining (open pit mining).	Treatment with acid.
Coal	Surface mining (open pit mining). Underground mining is now starting to become more common.	Coal is extracted in almost pure form.
Diamonds	Surface mining (alluvial deposits) and underground mining (pipe mining).	Diamonds are extracted from rocks and in almost pure form.
Copper	Surface mining (open pit mining).	Leaching is used to extract the copper using an acid.
Platinum	Underground mining (shaft mining).	Chemical methods and as a byproduct of copper mining.
Zinc	Underground mining (shaft mining).	Smelting and leaching.
Chromium (Chrome)	Surface mining (open pit mining) and underground mining (shaft mining).	Smelting, redox reactions.
Asbestos	Surface mining (open pit mining).	Extracted in fairly pure form.
Manganese	Surface mining (open pit mining) and underground mining (shaft mining).	Smelting and chemical processes.

Table 14.3: Table showing the mining and mineral extraction techniques for several minerals.

We will now discuss gold mining in more detail to gain a better understanding of the mining process.

Gold had long been known in Africa by the indigenous people, but in the late 1800's colonists found gold reefs and started exploiting the resource. Since then it has played a very important role (and often a controversial one too) in South Africa's *history* and *economy*. Its discovery brought many foreigners into South Africa, lured by the promises of wealth. They set up small mining villages which later grew into larger settlements, towns and cities. One of the first of these settlements was the beginning of present-day Johannesburg, also known as Egoli or Place of Gold.

Most of South Africa's gold is concentrated in the Golden Arc which stretches from Johannesburg to Welkom. Geologists believe that, millions of years ago, this area was a massive inland lake. Rivers feeding into this lake brought sand, silt, pebbles and fine particles of gold, depositing them over a long period of time. Eventually these deposits accumulated and became compacted to form gold-bearing sedimentary rock or **gold reefs**. It is because of this complex but unique set of circumstances that South Africa's gold deposits are so concentrated in that area. In other countries like Zimbabwe, gold occurs in smaller pockets which are scattered over a much greater area.

Mining the gold

ESBRK

A number of different techniques can be used to mine gold and other minerals. The three most common methods in South Africa are **panning**, **open pit** (not typically used for gold) and **shaft** mining.

1. Panning

Panning for gold is a manual technique that is used to sort gold from other sediments. Wide, shallow pans are filled with sand and gravel (often from river beds) that may contain gold. Water is added and the pans are shaken so that the gold is sorted from the rock and other materials. Because gold is much more dense, it settles on the bottom of the pan. **Pilgrim's Rest** in Mpumalanga was the first site for gold panning in South Africa.

2. Surface mining (open pit mining)

This type of mining takes place from the surface of the Earth. In open pit mining, the topsoil is removed first and placed on the side of a trench. Then the surface layers of rock are removed to expose the deeper, valuable mineral layers. The valuable rocks are then blasted into smaller rocks using explosives. The rocks are loaded onto huge trucks and taken away for further crushing and processing. Any mineral that is found close to the surface (even up to 1000 m) can be mined using surface mining techniques. If minerals are found deeper, as is the case with most of the gold in South Africa, underground mining is used.

3. Underground mining (shaft mining)

South Africa's thin but extensive gold reefs slope at an angle underneath the ground, and this means that some deposits are very deep and often difficult to reach. Shaft mining is needed to reach the gold ore. After the initial drilling, blasting and equipping of a mine shaft, tunnels are built leading outwards from the main shaft so that the gold reef can be reached. Shaft mining is a dangerous operation, and roof supports are needed so that the rock does not collapse. In addition the intense heat and high pressure below the surface make shaft mining very complex, dangerous and expensive. A diagram illustrating open cast and shaft mining is shown in Figure 14.3.

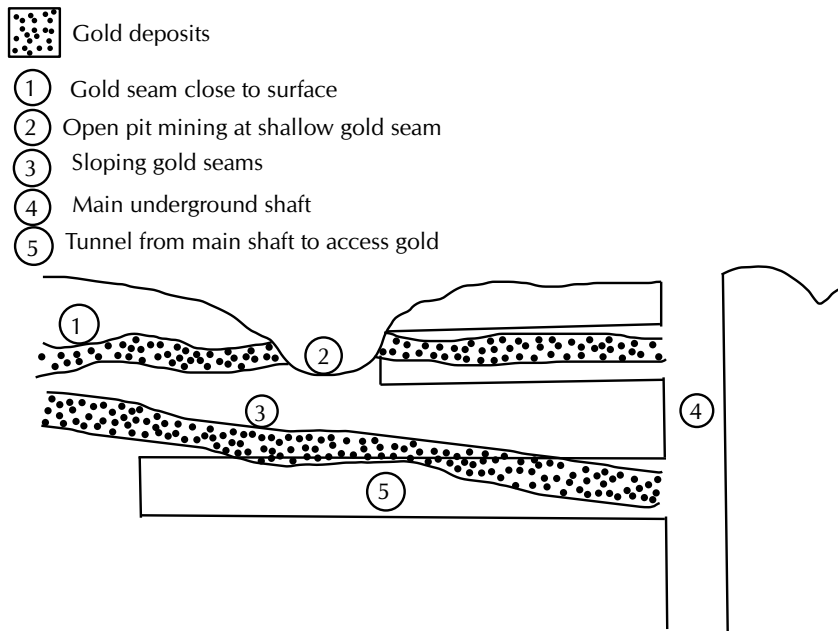


Figure 14.3: Diagram showing open pit and shaft mining.

Processing the gold ore

ESBRM

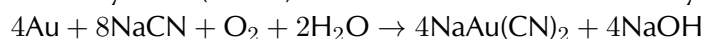
For every ton of ore that is mined, only a very small amount (about 5 g) of gold is extracted. A number of different methods can be used to separate gold from its ore, but one of the more common methods is called **gold cyanidation**.

In the process of gold cyanidation, the ore is crushed and then cyanide (CN^-) solution is added so that the gold particles are chemically separated from the ore. In this stage of the process, gold is oxidised. Zinc dust is then added to the cyanide solution. The zinc then takes the place of the gold, so that the gold is precipitated out of the solution. This process is shown in Figure 14.4.

STEP 1 - The ore is crushed until it is like a fine powder



STEP 2 - A sodium cyanide (NaCN) solution is mixed with the finely ground rock



Gold is **oxidised**.



STEP 3 - The gold-bearing solution can then be separated from the remaining solid ore through filtration.



STEP 4 - Zinc is added. Zinc replaces the gold in the gold-cyanide solution.

The gold is precipitated from the solution.

This is the **reduction** part of the reaction.

Figure 14.4: Flow diagram showing how gold is processed.

Gold has a number of uses because of its varied and unique characteristics. Below is a list of some of these characteristics that have made gold such a valuable metal:

- *Shiny*

Gold's beautiful appearance has made it one of the favourite metals for use in jewellery.

- *Durable*

Gold does not tarnish or corrode easily, and therefore does not deteriorate in quality. It is sometimes used in dentistry to make the crowns for teeth.

- *Malleable and ductile*

Gold can be bent and twisted into shape, as well as being flattened into very thin sheets. For this reason gold is used to make fine wires and thin, flat sheets.

- *Good conductor*

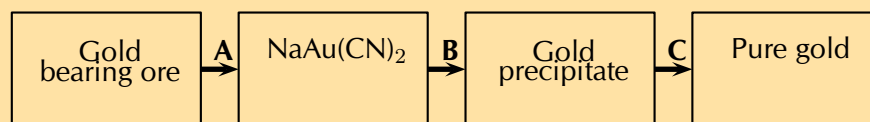
Gold is a good conductor of electricity and is therefore used in transistors, computer circuits and telephone exchanges.

- *Heat ray reflector*

Because gold reflects heat very effectively, it is used in space suits and in vehicles. It is also used in the protective outer coating of artificial satellites. One of the more unusual applications of gold is its use in firefighting, where a thin layer of gold is placed in the masks of the firefighters to protect them from the heat.

Exercise 14 – 1: Gold mining

1. In Mapungubwe (in the Limpopo Province) there is evidence of gold mining in South Africa as early as 1200. Today, South Africa is a world leader in the technology of gold mining. The following flow diagram illustrates some of the most important steps in the recovery of gold from its ore.



- a) Name the process indicated by A.
- b) During process A, gold is removed from the ore. Is gold oxidised or reduced during this process?
- c) Use oxidation numbers to explain your answer to the question above.
- d) Name the chemical substance that is used in process B.
- e) During smelting (illustrated by C in the diagram), gold is sent to a large oven called a furnace. Why do you think this process is needed, and explain what happens to the gold during this process.

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1. 247N



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Mining and the environment

ESBRP

Environmental impacts of gold mining

However, despite the incredible value of gold and its usefulness in a variety of applications, all mining is undertaken at a cost to the environment. The following are just a few of the environmental impacts of gold mining:

- *Resource consumption*

Gold mining consumes large amounts of electricity and water. Electricity is often generated using non-renewable resources.

- *Poisoned water*

If the mining process is not monitored properly, acid and other chemicals from gold processing can leach into nearby water systems such as rivers. This causes damage to animals and plants, as well as to humans who may rely on that water for drinking.

- *Changing the landscape*

This applies particularly to surface mines (open pit mines), where large amounts of soil and rock must be displaced in order to access the mineral reserves. The shape of the landscape can be changed when large amounts of rocks are dug up from the Earth and stacked on the surface. These are called mine dumps. Open pit mines also create very large holes (pits) in the ground that change the shape of the land.

- *Air pollution*

Dust from open pit mines, as well as harmful gases such as sulphur dioxide and nitrogen dioxide, could be released from mining processes and contribute to air pollution.

- *Threaten natural areas*

Mining activities often encroach on protected areas and threaten biodiversity in their operation areas.

Group discussion: Mine rehabilitation

There is a growing emphasis on the need to rehabilitate old mine sites that are no longer in use. If it is too difficult to restore the site to what it was before, then a new

type of land use might be decided for that area. Any mine rehabilitation programme should aim to achieve the following:

- ensure that the site is safe and stable
- remove pollutants that are contaminating the site
- restore the biodiversity that was there before mining started
- restore waterways to what they were before mining

There are different ways to achieve these goals. For example plants can be used to remove metals from polluted soils and water, and can also be used to stabilise the soil so that other vegetation can grow. Land contouring can help to restore drainage in the area.

In groups of 3-4, discuss the following questions:

1. What are the main benefits of mine rehabilitation?
2. What are some of the difficulties that may be experienced in trying to rehabilitate a mine site?
3. Suggest some creative ideas that could be used to encourage mining companies to rehabilitate old sites.
4. One rehabilitation project that has received a lot of publicity is the rehabilitation of dunes that were mined for titanium by Richards Bay Minerals (RBM). As a group, carry out your own research to try to find out more about this project.
 - What actions did RBM take to rehabilitate the dunes?
 - Was the project successful?
 - What were some of the challenges faced?

Group discussion: Mining and Mapungubwe

Read the following information about Mapungubwe.

Because of the history preserved at the site, Mapungubwe was declared a National Heritage Site in the 1980's and a World Heritage Site in 2003. The area surrounding it was declared a National Park in 1995. Unfortunately, the site is currently under huge threat from an Australian mining company who have been granted rights to construct an opencast and underground coal mine less than 6 km from the border of the National Park. There is much concern that mining operations will have a negative impact on the ecosystems, flora and fauna around Mapungubwe, and that it will hinder preservation of the site. Several environmental agencies have taken the matter to court in order to prohibit the planned mine and protect the environment around Mapungubwe.

As a class discuss, whether or not the proposed mining should go ahead. Divide into two teams. Assign one team to be in favour of the mining and one team against it. Each team should find facts relevant to their argument and then present it to the other team. At the end of the discussion, draw conclusions about whether or not the mining should take place.

Other minerals

ESBRQ

We have largely focused on the process involved in mining gold. However gold is not the only mineral of interest to humankind. Many other minerals are useful such as copper, iron and platinum.

Activity: Other mining activities

Choose one of the minerals given in the list below and find information on how that mineral is mined. Use the information given in this chapter about gold mining and try to find similar information for the mineral you have chosen. Write up your findings in a report. If possible, split the class into groups and assign a different mineral to each group.

- Iron
- Diamond
- Zinc
- Manganese
- Phosphate
- Copper
- Chrome
- Coal
- Platinum
- Asbestos

14.4 Energy resources

ESBRR

At present we rely mainly on oil and coal resources to provide our energy needs. However these resources are non-renewable energy sources and have many negative effects on the environment. Alternative sources of energy are constantly being investigated to assess if they will be able to provide for our growing energy needs. If we ignore these alternatives, one day we will run out of oil and coal reserves.

One of the most debated topics around energy resources is climate change. Scientists and environmentalists disagree as to the causes and effects of climate change and whether or not it actually exists.

Activity: Climate change

Climate change is a significant and lasting change in the distribution of weather patterns over periods of time ranging from decades to millennia. It is seen as changes in extreme weather events (freak storms, tornadoes, tsunamis, etc.) and changes in the average weather conditions for cities and regions.

Scientists collect data about climate change from ice core drilling, floral and faunal records, glacial and periglacial processes, isotopic analysis and from many other methods. Each of these methods provides different indicators of how the Earth's climate has changed over time.

Many factors influence the Earth's climate such as the amount of solar radiation received, changes in the Earth's orbit, changes on the Earth's surface, such as continental drift and size of polar ice caps, etc. All these factors combined make it difficult to pinpoint exactly what causes a particular change in climate. Also the effects of these events can take centuries to fully develop.

Human activities such as mining, burning of fossil fuels and agriculture also have an effect on the climate. Scientists are largely in agreement that the climate is changing irreversibly and that humans are a big cause of this change.

What is not agreed upon is how big an effect humans have and how much of a problem it is. At present global temperatures are found to be increasing and polar ice caps are found to be decreasing in size. Whether humans are simply making this worse or are actually causing the change is not yet understood.

Despite the debate, humans do need to be aware of their impact on the environment and will need to come up with ways to reduce it.

1. Discuss the above information.
2. Discuss ways in which human activities impact on the environment.
3. Discuss ways in which this impact can be reduced.
4. Conduct research on climate change and the debates about it. Critically analyse the information and draw your own conclusions. Discuss these ideas and conclusions with your classmates.

14.5 Summary

ESBRS

► See presentation: 247P at www.everythingscience.co.za

- The Earth's lithosphere is a rich source of many minerals.
- The lithosphere consists of the upper crust and mantle of the Earth.
- The minerals can be extracted using mining techniques.
- People have long been exploiting the Earth's mineral resources.
- Mining and mineral use has a significant environmental impact.
- Other resources in the lithosphere are fuels.

Exercise 14 – 2:

1. Give one word/term for each of the following descriptions:
 - a) The part of the Earth that includes the crust and in which all minerals are found.
 - b) The process in which minerals are extracted from the ores.
 - c) An age in which humans discovered the use of fire to improve the properties of stone.
2. Read the following extract and answer the questions that follow.

More profits, more poisons

Since the last three decades gold miners have made use of cyanidation to recover gold from the ore. Over 99% of gold from ore can be extracted in this way. It allows miners to obtain gold flakes – too small for the eye to see. Gold can also be extracted from the waste of old operations which sometimes left as much as a third of the gold behind.

The left-over cyanide can be re-used, but is more often stored in a pond behind a dam or even dumped directly into a local river. A teaspoonful of two-percent solution of cyanide can kill a human adult.

Mining companies insist that cyanide breaks down when exposed to sunlight and oxygen which render it harmless. They also point to scientific studies that show that cyanide swallowed by fish will not 'bio-accumulate', which means it does not pose a risk to anyone who eats the fish. In practise, cyanide solution that seeps into the ground will not break down because of the absence of sunlight. If the cyanide solution is very acidic, it could turn into cyanide gas which is toxic to fish. On the other hand, if the solution is alkaline, the cyanide does not break down.

There are no reported cases of human death from cyanide spills. If you don't see corpses, everything is okay.

- a) What is cyanidation?
- b) What type of chemical reaction takes place during this process: precipitation; acid-base; redox?
- c) Is the solution after cyanidation acidic, basic or neutral?
- d) How is the solid gold recovered from the solution?
- e) Refer to cyanidation and discuss the meaning of the heading of the extract: More profits, more poisons.

DBE exemplar paper - Nov 2007

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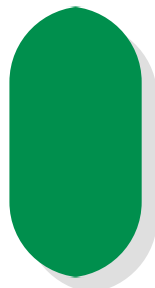


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Quantities used in the book



Physical Quantities		
Quantity	Unit name	Unit symbol
Acceleration (a)	metres per second squared	$\text{m}\cdot\text{s}^{-1}$
Charge (q)	coulomb	C
Concentration (C)	moles per cubic decimetre	$\text{mol}\cdot\text{dm}^{-3}$
Current (I)	amperes	A
Distance (d)	metre	m
Electrical energy (E)	joules	J
Electric field (E)	newtons per coulomb	$\text{N}\cdot\text{C}^{-1}$
Force (F)	newton	N
Induced emf ($\mathcal{E}$)	volt	V
Magnetic field (B)	tesla	T
Magnetic flux (ϕ)	weber	Wb
Mass (m)	kilogram	kg
Moles (n)	moles	mol
Pressure (p)	pascals	Pa
Power (P)	watts	W
Resistance (R)	ohms	Ω
Temperature (T)	kelvin	K
Tension (T)	newton	N
Time (t)	seconds	s
Voltage (V)	volts	V
Volume (V)	meters cubed	m^3
Weight (N)	newton	N

Table 15.1:
Units used in **the book**

1 Vectors in two dimensions

Exercise 1 – 2:

1. $\vec{R}_x = 1 \text{ N}$
2. $\vec{R}_x$ is 1,3 N and $\vec{R}_y = -1 \text{ N}$
3. $\vec{R}_x = 2 \text{ N}$ and $\vec{R}_y = 3 \text{ N}$
4. $\vec{R}_x = -2,5 \text{ N}$ and $\vec{R}_y = 2,5 \text{ N}$
5. $F_x = 2,8 \text{ N}$ and $F_y = -5 \text{ N}$

Exercise 1 – 5: Algebraic addition of vectors

1. 28,6 N at $53,53^\circ$
2. 25,4 N at $159,21^\circ$
3. 2,69 N at $328,67^\circ$
4. $186,4^\circ$

Exercise 1 – 6:

1.
 - $F_x = 3,54 \text{ N}$, $F_y = 3,54 \text{ N}$
 - $F_x = 6,81 \text{ N}$, $F_y = 13,37 \text{ N}$
 - $F_x = -6,80 \text{ N}$, $F_y = 9,02 \text{ N}$
 - $F_x = -52,83 \text{ N}$, $F_y = -113,29 \text{ N}$
2.
 - $F_x = 9,22 \times 10^4 \text{ N}$, $F_y = 5,99 \times 10^4 \text{ N}$
 - $F_x = 13,2 \text{ GN}$, $F_y = 7,04 \text{ GN}$
 - $F_x = -2,54 \text{ kN}$, $F_y = 11,01 \text{ kN}$
 - $F_x = 9,14 \times 10^6 \text{ N}$, $F_y = -8,52 \times 10^6 \text{ N}$

Exercise 1 – 7:

3. $\vec{R}_x = 1 \text{ N}$, $\vec{R}_y = 0 \text{ N}$.
4. $\vec{R}_x = 1,6 \text{ N}$, $\vec{R}_y = 0,2 \text{ N}$
5. $\vec{R}_x = 1,4 \text{ N}$, $\vec{R}_y = 1,7 \text{ N}$
12. 3,25 N at 45°
13.
 - a) $F_x = 96,29 \text{ N}$ and $F_y = 41,87 \text{ N}$
 - b) $F_x = 13,95 \text{ N}$ and $F_y = 23,12 \text{ N}$
 - c) $F_x = 9,02 \text{ N}$ and $F_y = -6,8 \text{ N}$
 - d) $F_x = -62,97 \text{ N}$ and $F_y = -135,04 \text{ N}$
 - e) $F_x = 3,89 \text{ N}$ and $F_y = 14,49 \text{ N}$
 - f) $F_x = 14,43 \text{ N}$ and $F_y = 3,71 \text{ N}$
 - g) $F_x = -6,22 \text{ N}$ and $F_y = 9,43 \text{ N}$
 - h) $F_x = -136,72 \text{ N}$ and $F_y = 99,34 \text{ N}$

- | | |
|--|----------------------|
| 14. have equal magnitudes but opposite directions. | 17. 4 N |
| 15. 15 N due north and 7 N due west | 18. 902,98 N |
| 16. 0° | 19. a) B
b) 781 N |

2 Newton's laws

Exercise 2 – 1:

- | | |
|---|---|
| 1. Kinetic friction = 0 N and static friction = 20 N to the left. | d) 4 N |
| 2. a) 0 N for both static and kinetic frictional forces. | e) Kinetic friction: 4 N and static friction: 0 N |
| b) Kinetic friction: 0 N and static friction: 5 N | 3. a) |
| c) 8 N | $F_s \leq \mu_s N$ |

Exercise 2 – 2:

- | | |
|--|------------------|
| 1. $\vec{F}_{gy} = 172,30$ N in the negative y -direction, $\vec{F}_{gx} = 422,20$ N in the negative x -direction. | 2. $42,61^\circ$ |
|--|------------------|

Exercise 2 – 3: Forces and motion

- | | |
|---------------------------------------|------------------------------|
| 1. c) $\vec{F}_R = 55$ N to the right | 2. c) $\vec{F}_f = 346,41$ N |
|---------------------------------------|------------------------------|

Exercise 2 – 5:

- | | |
|--------------------------------------|---------------------------------------|
| 2. 1700 N | 5. 17,5 N |
| 3. 12 N | 6. 20 kg |
| 4. $0,1 \text{ m}\cdot\text{s}^{-2}$ | 7. $0,15 \text{ m}\cdot\text{s}^{-2}$ |

- | | |
|--|---|
| 8. 8,33 kg | 16. 2500 N |
| 9. $0,25 \text{ m}\cdot\text{s}^{-2}$ and $1 \text{ m}\cdot\text{s}^{-2}$ | 17. 6 N |
| 10. a) 40 N
b) $20 \text{ m}\cdot\text{s}^{-2}$ | 18. a) $10,2 \text{ m}\cdot\text{s}^{-2}$ |
| 11. a) $5 \text{ m}\cdot\text{s}^{-2}$
b) $100 \text{ m}\cdot\text{s}^{-1}$
c) 7 s
d) 562,5 m | 19. a) $-10 \text{ m}\cdot\text{s}^{-2}$
b) 10 000 N |
| 12. a) 100 N
b) -25 N
c) 173,2 N | 20. 277,76 N, 113,35 N and 45° |
| 13. 8,5 N | 21. $25,53^\circ$ |
| 14. $4,2 \text{ m}\cdot\text{s}^{-2}$ | 22. $25,53^\circ$ |
| | 23. a) -1340 N
b) 562,5 N |
| | 24. a) $-441,14 \text{ N}$
b) 20,97 N |

Exercise 2 – 6:

- | | |
|-------------|-----------|
| 1. the same | 2. A or B |
|-------------|-----------|

Exercise 2 – 7:

- | | |
|--|--------------------------------------|
| 1. $1,92 \times 10^{18} \text{ N}$ | 5. a) $7 \times 10^{-7} \text{ N}$ |
| 2. $8,80 \times 10^{17} \text{ N}$ | b) $1,35 \times 10^{-6} \text{ N}$ |
| 3. Earth: $5,65 \times 10^6 \text{ m}$, Jupiter: $1,00 \times 10^8 \text{ m}$ | c) 0,52 |
| 4. Jupiter: $24,8 \text{ m}\cdot\text{s}^{-2}$ Moon: $1,7 \text{ m}\cdot\text{s}^{-2}$ | 6. a) $4,7 \times 10^{12} \text{ N}$ |
| | b) $9,4 \times 10^{16} \text{ N}$ |

Exercise 2 – 8:

- | | |
|-------------------------------|-------------|
| 1. a) 323,75 N
b) 326,04 N | c) 980,88 N |
| | 2. Object 2 |

Exercise 2 – 9:

- | | |
|-----------------------------|--|
| 1. 5 F | 8. 0,13 m |
| 2. decrease | 9. 0,5 kg |
| 3. $r - R = 2R - R = R$ | 10. a) Newton's law of universal gravitation |
| 4. $\frac{1}{9}$ | b) It is the same |
| 5. 2 M | c) increases |
| 7. $6,67 \times 10^{-12}$ N | d) $2,37 \times 10^7$ N |

Exercise 2 – 10: Forces and Newton's Laws

- | | |
|--|--|
| 2. a) non-contact
b) contact
c) contact
d) non-contact | 24. It accelerates to the left, moving along the smooth horizontal surface. |
| 4. acceleration | 25. A zero resultant force acts on the passenger. |
| 5. 2 N | 26. b) 696 000 N
c) 2784 N |
| 6. 250 | 27. b) i. $5 \text{ m}\cdot\text{s}^{-2}$
ii. 5000 N
c) 2500 N |
| 7. The box moves to the left. | 29. a) 11 760 N
b) $-2,25 \text{ m}\cdot\text{s}^{-2}$
c) 9060 N |
| 8. 200 | 30. a) $1,33 \text{ m}\cdot\text{s}^{-2}$
b) $26,7 \text{ m}\cdot\text{s}^{-1}$
c) $1,25 \text{ m}\cdot\text{s}^{-2}$
d) $51,6 \text{ m}\cdot\text{s}^{-1}$ |
| 9. $F\cos 60^\circ - ma$ in the direction of A | 31. b) $W_\perp = 16,86 \text{ N}$ and $W_\parallel = 24,01 \text{ N}$ |
| 10. his inertia | 32. c) 1171,28 kg |
| 12. The resultant of the forces is zero. | 33. b) $0,33 \text{ m}\cdot\text{s}^{-2}$
c) $F_f = 300 \text{ N}$
e) 50 m |
| 13. F | 34. a) $-6,67 \text{ m}\cdot\text{s}^{-2}$
c) 3,0 s
d) 2 s |
| 14. F is the reaction to the force that the rocket exerts on the gases which escape out through the tail nozzle. | |
| 15. move to the left | |
| 16. b) 19,6, 49 | |
| 17. increase only | |
| 18. graph (b) | |
| 19. 40 N | |
| 21. $\frac{5}{4}a$ | |
| 22. There is zero resultant force. | |
| 23. The load has a weight equal in magnitude to F. | |

35. $\sin \theta = \frac{N}{w}$ ii. 204,17 kg
36. Force F_3 is the resultant of forces F_1 and F_2 .
37. c) 1,63 N
38. b) i. 5497,2 N
39. a) 2352 N
b) 1568 N
c) 2715,9 N
d) 1810,6 N

Exercise 2 – 11: Gravitation

1. $9F$
2. weight of the object
3. a) 778,2 N
b) 3112,8 N
4. $8,2 \times 10^{-2}$ m
5. a) 448 N
b) 192 N
6. 5488 N
7. $1,6 \times 10^6$ N
9. $4,1 \times 10^{22}$ N
11. a) Agree
b) Disagree
c) Disagree
12. a) 3000 N
b) 76,53 kg
13. 1,8 N
14. b) 1,63
c) on the moon

3 Atomic combinations

Exercise 3 – 5: Atomic bonding and Lewis diagrams

3. a) Nitrogen: 5, hydrogen: 1
Carbon: 4, hydrogen: 1
c) ammonia and methane
4. a) 6
- b) 1
- c) 2 single bonds
- d) Y: oxygen and X: hydrogen.

Exercise 3 – 6: Molecular shape

1. linear
2. linear
3. trigonal pyramidal
4. octahedral
5. linear
6. tetrahedral
7. bent or angular
8. trigonal planar

Exercise 3 – 7:

1. 1,0, 0,4, 3,0, 0,6 and 1,0

Exercise 3 – 8: Electronegativity

1. a) 2,1 will have a slightly negative charge.
b) 3,0
c) Hydrogen will have a slightly positive charge and chlorine
d) Polar covalent bond
e) polar

Exercise 3 – 9:

1. a) Bond length b) linear
b) Covalent bond c) trigonal planar
c) Electronegativity d) linear
2. Dative covalent (co-ordinate covalent) bond e) tetrahedral
f) tetrahedral
4. $\text{Ne} + \text{Ne} \rightarrow \text{Ne}_2$ g) linear
h) octahedral
6. a) 1
b) 5 9. a) non-polar
c) X could be hydrogen and Y could be nitrogen. b) non-polar
c) non-polar
7. a) linear d) polar

4 Intermolecular forces

Exercise 4 – 1:

1. dipole-dipole 3. ion-dipole
2. induced dipole 4. induced dipole

Exercise 4 – 4:

- | | |
|---|-----------------------------------|
| 1. a) Intermolecular force | higher than for Cl ₂ |
| b) Polar molecule | |
| c) Specific heat | 3. a) HI and NH ₃ only |
| 2. The melting point of NH ₃ will be | 4. a) Water |

5 Geometrical optics**Exercise 5 – 1: Rays and Reflection**

- | | |
|---------|---------|
| 3. a) E | 7. 15° |
| b) C | |
| c) D | 8. 45° |
| d) B | |
| e) A | 9. 65° |
| 6. B | 10. 25° |

Exercise 5 – 2: Refractive index

- | | |
|---|----------------|
| 1. $2,29 \times 10^8 \text{ m}\cdot\text{s}^{-1}$ | 2. crown glass |
|---|----------------|

Exercise 5 – 4: Snell's Law

- | | |
|-------------------------|---|
| 2. b) 25,88° | 7. 1,327 |
| 3. 18,08° | 8. 30,34° |
| 4. a) decreases | 9. 78,07° |
| b) remains the same | |
| c) Towards the normal. | 10. A: acetone, B: fused quartz, C: diamond, D: cubic zirconia, E: 80% sugar solution |
| 5. a) increases | |
| b) remains the same | |
| c) Away from the normal | 11. c) water |

Exercise 5 – 5: Total internal reflection and fibre optics

- | | |
|--|-------------------------------|
| 3. No | 7. a glass to water interface |
| 4. no | 8. $41,8^\circ$ |
| 5. $47,33^\circ$ | 9. $62,46^\circ$ |
| 6. The light will be trapped in the diamond. | 10. $2,42^\circ$ |
| | 11. $37,86^\circ$ |

Exercise 5 – 6: End of chapter exercises

- | | |
|---------------|---------------------|
| 1. a) normal | 4. 35° |
| b) refraction | |
| c) reflection | 5. b) $27,78^\circ$ |
| | c) increases |
| 2. a) False | d) stays the same |
| b) True | |
| c) True | 6. sapphire |

6 2D and 3D wavefronts

Exercise 6 – 1:

2. light waves

Exercise 6 – 2:

- | | |
|-----------------------------|--------------------------------------|
| 2. blue, green, yellow, red | 4. a) $3,4 \times 10^{-6} \text{ m}$ |
| | b) $2,4 \times 10^{-6} \text{ m}$ |

7 Ideal gases

Exercise 7 – 2: Boyle's law

- | | |
|------------------------|--------------|
| 1. 100 kPa | 4. c) Yes |
| 2. 112,5 kPa | |
| 3. $8,33 \text{ cm}^3$ | 5. a) 217 Pa |

Exercise 7 – 3: Charles' law

- | | |
|-------------------------|------------|
| 2. 4,67 dm ³ | 3. 56,67 K |
|-------------------------|------------|

Exercise 7 – 4: Pressure-temperature relation

- | | |
|---------------------------|-------------|
| 1. c) linear relationship | 3. 8,14 atm |
| 2. 7,24 atm | 5. 465,6 K |

Exercise 7 – 5: The general gas equation

- | | |
|-------------|-------------|
| 1. 0,87 atm | 3. 1,26 atm |
| 2. 0,6 atm | |

Exercise 7 – 6: The ideal gas equation

- | | |
|-------------------------|--------------|
| 1. 0,008 m ³ | 4. b) No |
| 2. 106,54 kPa | c) 193,3 kPa |
| 3. 0,02 mol | 5. a) 73,1 g |

Exercise 7 – 7:

- | | |
|---------------------------------------|---|
| 1. a) Ideal gas | 3. The sample of helium gas will be at the greatest pressure. |
| b) Charles' law | 4. $V \propto n$, with p, T constant. |
| c) Temperature | 7. b) 311,775 Pa |
| 2. Temperature must be kept constant. | 8. 106,39 kPa |

- | | |
|-----------------|----------------|
| 9. c) 322,0 kPa | 12. a) 201,9 K |
| d) 246,31 g | b) Decreased. |
| 10. 1363,0 kPa | 13. d) 64 kPa |
| 11. 5,78 L | e) Increases. |

8 Quantitative aspects of chemical change

Exercise 8 – 1:

1. 2 dm³

Exercise 8 – 2: Gases and solutions

- | | |
|---------------------------------|-------------------------------|
| 1. 7 dm ³ | 3. 1,105 mol·dm ⁻³ |
| 2. a) 4,56 mol·dm ⁻³ | 4. 0,24 mol·dm ⁻³ |
| b) 60,95 g | 5. 0,185 mol·dm ⁻³ |

Exercise 8 – 3:

1. 2,74 kg

Exercise 8 – 4:

1. 65,69%

Exercise 8 – 5:

1. C₂H₂O₄

Exercise 8 – 6:

1. 69%

Exercise 8 – 7: Stoichiometry

- | | |
|----------------------------------|----------|
| 1. 1,69 kg | 4. 93,5% |
| 2. 88,69% | |
| 3. C ₆ H ₆ | 5. 48,6% |

Exercise 8 – 8: Gases 2

2. 0,224 dm³

Exercise 8 – 9:

- | | |
|---|---|
| 1. a) Limiting reagent | 7. 1,52 mol·dm ⁻³ |
| b) Empirical formula | 8. a) O ₃ : 0,0154 mol and NO:
0,0223 mol |
| c) Theoretical yield | b) O ₃ |
| d) Titration | c) 0,71 g |
| 2. 67,2 dm ³ | 9. 96,9% |
| 3. 1,8 dm ³ | 10. a) 88,461 g |
| 4. 0,1 mol·dm ⁻³ | b) 66,051 dm ³ |
| 5. 58% | c) 77,69 g |
| 6. a) 2NaOH + H ₂ SO ₄ → Na ₂ SO ₄ +
2H ₂ O | 11. C ₂ Cl ₂ F ₄ |
| b) 0,03 mol | 12. 48,6% |
| c) the sodium hydroxide will be
fully neutralised | 13. a) 1,33 dm ³ |
| | b) Yes |

9 Electrostatics

Exercise 9 – 1: Electrostatic forces

- | | |
|------------------------------------|---|
| 1. $1,35 \times 10^{-2} \text{ N}$ | 8. $8,77 \text{ N}$ to the right. |
| 2. $1,27 \times 10^{-3} \text{ N}$ | 9. $4,2 \times 10^{-8} \text{ C}$ |
| 3. $0,26 \text{ N}$ | 10. $3,42 \times 10^{-5} \text{ N}$ acting at an angle of $83,8^\circ$ to the negative x -axis |
| 4. $0,56 \text{ N}$ | 11. $1,52 \times 10^{-6} \text{ N}$ acting at an angle of $82,80^\circ$ to the positive x -axis |
| 5. $0,2$ | 12. $1,05 \times 10^{-4} \text{ N}$ acting at an angle of $34,76^\circ$ to the positive x -axis |
| 6. 9000 N | |
| 7. $4,6 \times 10^{-3} \text{ m}$ | |

Exercise 9 – 2: Electric fields

- | | |
|---------------------------------------|--|
| 1. $0,15 \text{ N}\cdot\text{C}^{-1}$ | 2. $-5,8 \times 10^{-8} \text{ N}\cdot\text{C}^{-1}$ |
|---------------------------------------|--|

Exercise 9 – 3:

- | | |
|---|--|
| 1. $8,44 \times 10^{-7} \text{ N}\cdot\text{C}^{-1}$ | 5. $\frac{1}{3}F$ |
| 2. a) $1,35 \times 10^{-3} \text{ N}\cdot\text{C}^{-1}$ | 6. a |
| b) -2 nC | 7. b) $1,5 \times 10^{-5} \text{ N}\cdot\text{C}^{-1}$ |
| c) $1,00 \times 10^{-5} \text{ N}\cdot\text{C}^{-1}$ | 8. $2E$ |
| 3. $3,3 \times 10^{-4} \text{ m}$ | |

10 Electromagnetism

Exercise 10 – 1: Magnetic Fields

- | | |
|-----------------------|--|
| 3. a) Out of the page | 4. A - D: counterclockwise, E - H: clockwise |
| b) into the page | |

Exercise 10 – 2: Faraday's Law

- | | |
|---------------|---------------|
| 5. 2,45 V | 7. a) 28,8 Wb |
| 6. a) -0,14 V | b) 9,22 s |
| b) 0,22 m | |

Exercise 10 – 3:

- | | |
|--|-----------------|
| 3. a) 0,1225 Wb | 6. 9,34 V |
| b) The change in flux is 0,117 Wb and the induced emf will be zero because the cardboard is not a conductor. | 7. a) 0,00249 V |
| | b) 0,014 m |
| 4. 3,4 Wb and 0 Wb | 8. 0,012 Wb |
| 5. 7,26 Wb | 9. 2,65 V |

11 Electric circuits

Exercise 11 – 1: Ohm's Law

- | | |
|---------------------|--------|
| 1. b) straight-line | d) Yes |
| c) 0,13 | |

Exercise 11 – 2: Ohm's Law

- | | |
|---------------|---------|
| 1. 4 Ω | 3. 15 V |
| 2. 3 A | |

Exercise 11 – 3: Series and parallel resistance

- | | |
|--------------------|--|
| 1. 20 k Ω | 4. 6 Ω |
| 2. 90 Ω | |
| 3. 9,09 k Ω | 5. a) 1,2 Ω b) 0,48 Ω c) 5 Ω d) 10 Ω |

Exercise 11 – 4: Ohm's Law in series and parallel circuits

1. $2\ \Omega$
2. $1,8\ \text{A}$
3. $0,75\ \text{A}$
4. $12\ \text{A}$
5. $30\ \Omega$
6. a) $26\ \text{A}$
b) $R_1 = 20\ \text{A}, R_2 = 4\ \text{A}, R_3 = 2\ \text{A}$

Exercise 11 – 5: Series and parallel networks

1. a) $3,33\ \Omega$
b) $10,67\ \Omega$
c) $2,652\ \Omega$
2. a) $4,52\ \text{A}$
b) $0,59\ \text{A}$
3. Total voltage: $12\ \text{V}, R_1 = 9,32\ \text{V}, R_2 = 2,66\ \text{V}, R_3 = 2,66\ \text{V},$
4. a) $2,5\ \text{A}$
b) $3,75\ \text{V}$
c) $1,25\ \text{A}$

Exercise 11 – 6:

1. $2,00 \times 10^{12}\ \text{W}$
2. $0,99\ \text{W}$
3. a) $1,5\ \text{W}$
b) $7,5\ \text{W}$
4. a) $6,48\ \text{W}, 1,44\ \text{W}, 2,88\ \text{W}$
b) $1,88\ \text{W}, 1,25\ \text{W}, 0,16\ \text{W}, 0,32\ \text{W}$
- c) $30,2\ \Omega, 0,89\ \text{W}, 1,17\ \text{W}, 5,83\ \text{W}, 3,5\ \text{W}$
5. a) $2,64\ \text{A}$
b) $0,34\ \text{A}$
c) $1,7\ \text{W}$
6. $18,64\ \text{W}, 10,72\ \text{W}, 5,36\ \text{W}$

Exercise 11 – 7:

1. a) Voltage
b) Electrical power
c) Ohm's law
2. $0,5\ \text{A}$
3. $17\ \Omega$
4. $1,05\ \Omega$
5. $24\ \text{W}$
6. $0,5\ \text{A}, 1,5\ \text{V}, 5\ \text{V}, 2,5\ \text{V}$
7. a) $0,09\ \text{A}$
b) $2,25\ \text{V}$
c) $13,5\ \text{V}$
8. a) $38,4\ \Omega$
b) $270\ \text{kJ}$
9. b) $-1,014 \times 10^{-16}\ \text{J}$

12 Energy and chemical change

Exercise 12 – 1: Exothermic and endothermic reactions 1

- | | |
|----------------|------------------|
| 1. a) taken in | 2. a) Exothermic |
| b) released | b) Endothermic |
| c) released | c) Exothermic |
| d) taken in | d) Endothermic |

Exercise 12 – 2: Endothermic and exothermic reactions

- | | |
|-------------------|----------------|
| 1. a) Exothermic | b) Exothermic |
| b) Exothermic | c) Exothermic |
| c) Endothermic | d) Endothermic |
| d) Exothermic | e) Exothermic |
| 2. a) Endothermic | f) Exothermic |

Exercise 12 – 3: Energy and reactions

- | | |
|----------------|----------|
| 1. Endothermic | b) 0 kJ |
| | c) 15 kJ |
| 2. a) –15 kJ | d) 40 kJ |

Exercise 12 – 4:

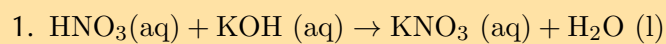
- | | |
|--|--|
| 1. a) Activation energy | 3. The energy of the product is less than the energy of the reactants. |
| b) Bond energy | 4. The second graph (b) is correct. |
| c) Exothermic reaction | 6. a) Negative |
| d) Endothermic reaction | c) Photosynthesis |
| 2. Energy is released when the bonds in H ₂ O form. | d) Endothermic |

13 Types of reactions

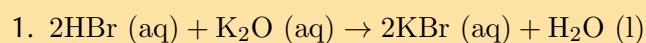
Exercise 13 – 1: Acids and bases

2. c) Amphoteric

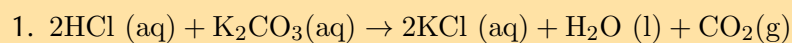
Exercise 13 – 3:



Exercise 13 – 4:



Exercise 13 – 5:

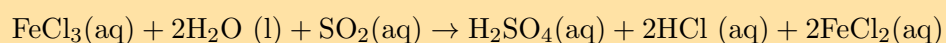


Exercise 13 – 7: Oxidation numbers

- | | |
|-----------------------------------|--|
| 1. a) Magnesium: +2, fluorine: -1 | c) Hydrogen: +1, carbon: -4 |
| b) Calcium: +2, chlorine: -1 | d) Magnesium: +2, oxygen: -2, sulfur: +6 |

Exercise 13 – 8: Redox reactions

1. Fe is oxidised and O_2 is reduced
- 2.



Exercise 13 – 9:

- | | |
|--|--|
| 1. a) Redox reaction | 6. $2\text{HCl (aq)} + \text{Mg(OH)}_2\text{(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + 2\text{H}_2\text{O (l)}$ |
| b) Bronsted-Lowry base | |
| c) Oxidation | 7. $2\text{HCl (aq)} + \text{Na}_2\text{CO}_3\text{(aq)} \rightarrow 2\text{NaCl (aq)} + \text{H}_2\text{O (l)} + \text{CO}_2\text{(g)}$ |
| d) Amphoteric substance | |
| 2. HNO_3 donates protons and is a Bronsted-Lowry acid | 8. $2\text{H}_3\text{PO}_4\text{(aq)} + 3\text{CaO (s)} \rightarrow \text{Ca}_3(\text{PO}_4)_2\text{(aq)} + 3\text{H}_2\text{O (l)}$ |
| 3. Br^- is the oxidising agent (chlorine is reduced in this reaction) | 10. a) i. -1
ii. -2
iii. 0 |
| 4. H_2SO_3 is the Bronsted-Lowry acid and KOH is the Bronsted-Lowry base. | 13. $\text{Cd (s)} + 2\text{NiO(OH) (s)} + 2\text{H}_2\text{O (l)} \rightarrow \text{Cd(OH)}_2\text{(s)} + 2\text{Ni(OH)}_2\text{(s)} + 2\text{OH}^-\text{(aq)}$ |

14 The lithosphere

Exercise 14 – 1: Gold mining

- | | |
|-------------------------|---------|
| 1. b) Gold is oxidised. | d) Zinc |
|-------------------------|---------|

Exercise 14 – 2:

- | | |
|--------------------------------------|-----------------------------------|
| 1. a) Lithosphere | c) Middle Stone Age |
| b) Minerals processing or extraction | 2. b) Redox reaction
c) acidic |

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